

CHALLENGES AND OPPORTUNITIES OF MISSION- ORIENTED INNOVATION POLICY IN KOREA

OECD SCIENCE, TECHNOLOGY
AND INDUSTRY
POLICY PAPERS

March 2025 **No. 172**

OECD Science, Technology and Industry Policy Papers

This paper was approved and declassified by written procedure by the Committee for Scientific and Technological Policy (CSTP) on 19/02/2025 and prepared for publication by the OECD Secretariat. Any finding of this work is without prejudice to any reviews that may be conducted by investigating authorities or under the WTO dispute settlement procedures on subsidies and countervailing measures.

Note to Delegations:

This document is also available on O.N.E Members & Partners under the reference code:

DSTI/STP(2024)32/FINAL

This document, as well as any data and map included herein, are without prejudice to the status of or sovereignty over any territory, to the delimitation of international frontiers and boundaries and to the name of any territory, city or area.

© OECD 2025



Attribution 4.0 International (CC BY 4.0)

This work is made available under the Creative Commons Attribution 4.0 International licence. By using this work, you accept to be bound by the terms of this licence (<https://creativecommons.org/licenses/by/4.0/>).

Attribution – you must cite the work.

Translations – you must cite the original work, identify changes to the original and add the following text: *In the event of any discrepancy between the original work and the translation, only the text of original work should be considered valid.*

Adaptations – you must cite the original work and add the following text: *This is an adaptation of an original work by the OECD. The opinions expressed and arguments employed in this adaptation should not be reported as representing the official views of the OECD or of its Member countries.*

Third-party material – the licence does not apply to third-party material in the work. If using such material, you are responsible for obtaining permission from the third party and for any claims of infringement.

You must not use the OECD logo, visual identity or cover image without express permission or suggest the OECD endorses your use of the work.

Any dispute arising under this licence shall be settled by arbitration in accordance with the Permanent Court of Arbitration (PCA) Arbitration Rules 2012. The seat of arbitration shall be Paris (France). The number of arbitrators shall be one.

Abstract

This study provides a comprehensive understanding of how Korea implements mission-oriented innovation policies (MOIPs), and offers guidance on how to design, co-ordinate, and implement MOIPs in the country.

As part of a series of MOIP national reviews, the paper finds that Korea has made significant efforts to embrace a mission-oriented approach, setting several overarching MOIP frameworks and specific MOIP initiatives. This work builds upon a long history of central co-ordination of STI policies implemented in a number of STI and sectoral public bodies. MOIPs in Korea are the latest attempt to provide STI with more direction in order to prioritise technological development that both benefits Korean society and drive competitiveness. The study concludes with options for change, including by simplifying prioritisation exercises for sectors and technologies across government, and exploring how the government can better support its societal missions to drive more transformative outcomes.

Acknowledgements

This policy paper has been prepared and drafted by Charles McIvor and Philippe Larrue, who work within the OECD Directorate for Science Technology and Innovation (STI), Science and Technology (STP) division. This work is conducted under the aegis of the OECD Committee for Scientific and Technological Policy (CSTP) during the biennium 2023-2024.

This study was commissioned by the Science and Technology Policy Institute (STEPI), Korea.

We wish to acknowledge the contributions of Dr Sang-Min Park, Director, R&D Investment Planning Division, STI Office, Dr Byeongwon Park, Researcher, STEPI and Yongsuk Jang, Research Fellow, STEPI and Chair of the CSTP. Their support during the fieldwork, drafting and consultations was essential in helping undertake this study.

Blandine Serve provided valuable statistical support (Science and Technology Policy division, OECD) and Yeonhwa Kim (also from Science and Technology Policy division at the time) contributed to the literature review.

Detailed information on the Comprehensive Plan to Solve Social Problems Based on Science and Technology, National Strategic Technology Policy, and the Alchemist are available on the MOIP Online Toolkit (<https://stip.oecd.org/stip/moip>), a comprehensive knowledge platform on mission-oriented innovation policies. The development of the MOIP Online Toolkit was funded by the European Union's Horizon 2020 research and innovation programme and its recent update by Horizon Europe.

Drafts of the paper went through several rounds of fact-checking. The authors would like to thank all those who provided comments and inputs during this process of validation, including Dr Jin-Yong Kim, Director, Centre for National Strategic Technology Policy, Korea Institute of S&T Evaluation and Planning; Jinhee Park, Director, Performance Evaluation Policy Division, STI Office; Dr Gwi-Nam Bae, Head / Principal Research Scientist, Center for FRIEND Project, Korea Institute of Science and Technology; Yongseong Kang, Senior Project Researcher, Challenge & Innovation Team, Korea Evaluation Institute of Industrial Technology; Sojung You, International Cooperation Specialist, Climate and Air Quality Policy Office, Ministry of Environment; and Dr Jeehoon KI, Associate Research Fellow, Korea Institute of S&T Evaluation and Planning. These participants are not accountable for the material in this paper.

Finally, the project has benefited from the information and analysis in the OECD Reviews of Innovation Policy: Korea 2023 (<https://doi.org/10.1787/bdcf9685-en>).

The authors are grateful to Kyriakos Vogiatzis (DSTI) for his secretarial assistance and to Sylvain Fraccola (DSTI) for his communications and publication support.

Table of contents

Abstract	3
Acknowledgements	4
Executive summary	8
1 Introduction	10
2 Mission-oriented innovation policy review methodology	12
The definition of MOIPs	12
The three dimensions of mission-orientation	13
Types of mission-oriented innovation policies	13
3 Framework conditions for mission-orientation in Korea	15
Strategic orientation	16
Policy co-ordination	19
Policy implementation	21
4 Mission-oriented policy initiatives in practice	23
The Comprehensive Plan to Solve Social Problems Based on Science and Technology	23
National Strategic Technology Policy	28
The Alchemist	34
5 Conclusions and options for the future of the mission-oriented policy agenda in Korea	38
Korea's STI institutional setting is conducive to mission-oriented policies	38
Options for change	39
References	42
Annex A. Detailed framework conditions for mission-orientation in Korea	47
The development of the Korean STI governance system	47
Strategic Orientation	48
Policy co-ordination	57
STI Policy implementation	60

Annex B. Overview of the process, governance and main MOIP features of the three cases	64
Annex C. 3rd Comprehensive Plan to Solve Social Problems Based on Science and Technology – List of social problems	66
Annex D. 3rd Comprehensive Plan to Solve Social Problems Based on Science and Technology – core problems	68
Annex E. Reducing fine particle matter	69
Annex F. National Strategic Technology Policy – strategic technologies	71
Annex G. The Alchemist – programme themes	73
Annex H. The Innovation Challenge Programme	75
Annex I. Main features of selected MOIP initiatives in Korea	77
Annex J. List of interviewees	79
Notes	80

Tables

Table 2.1. The three dimensions of mission-orientation	13
Table 2.2. Basic characteristics of the four main types of MOIP	13
Table 3.1. The three dimensions of mission-orientation	15
Table 3.2. Korean models for competitive R&D programmes	21
Table A.1. Internal structure and functions of the Science, Technology and Innovation Office	58
Table C.1. 3rd Comprehensive Plan to Solve Social Problems Based on Science and Technology – List of social problems	66
Table D.1. 3rd Comprehensive Plan to Solve Social Problems Based on Science and Technology – core problems	68
Table F.1. National Strategic Technology Policy – strategic technologies	71
Table G.1. The Alchemist – programme themes	73
Table I.1. Annex I. Main features of selected MOIP initiatives in Korea	77
Table J.1. List of interviewees	79

Figures

Figure 1.1. Structure of the mission-oriented innovation policy review	11
Figure 2.1. The definition of MOIPs and the three MOIP dimensions	13
Figure 3.1. Korea's STI governance structure	19
Figure 4.1. Elderly life safety roadmap	25

Figure A.1. Korea's STI governance system: From long-term strategic orientations to annual programming and budgeting	49
Figure A.2. Overview of Korea's annual budgeting and programming process	51
Figure A.3. Korea's different types of evaluation by programme	53
Figure A.4. Gross domestic expenditures on R&D (GERD), (2022 and 2013)	54
Figure A.5. Government Budget Allocations for R&D (GBARD), (2022 and 2012)	55
Figure A.6. Oriented and non-oriented civil GBARD, (2022)	55
Figure A.7. R&D budgets by SDG-related category, (2021 or latest available data)	56
Figure A.8. Direct government funding and government tax support for business R&D (2021 or nearest year and 2006 or closest available data)	61
Figure B.1. The 3rd comprehensive plan to solve social problems based on science and technology (2023-27)	64
Figure B.2. The National Strategic Technology Policy	65
Figure B.3. The Alchemist	65

Boxes

Box 3.1. Korean green strategies	16
Box 3.2. France's Acceleration Strategies – an example of horizontal budgeting	18
Box 4.1. Procurement for innovation in Korea	27
Box 4.2. National Strategic Technology – example of mission framing	30
Box 4.3. Co-ordinating missions with structural funding	32
Box 4.4. The Carbon-Neutral Strategic Technology Policy	33
Box 4.5. Overarching MOIPs using challenge-oriented policies in their policy mix	36

Executive summary

This review, undertaken as part of the OECD Committee for Scientific and Technological Policy (CSTP) work on mission-oriented innovation policies (MOIPs), aims to i) identify the key elements that foster or hinder the development and implementation of MOIPs to tackle societal challenges in Korea; ii) review past and on-going Korean MOIPs; iii) provide evidence-based conclusions and options for improving the adoption and implementation of this policy approach in the future, building on the Korean STI system's distinctive strengths, and learning from past mission experiences in Korea and internationally.

Does Korea have specific features that are conducive to the design and implementation of MOIPs?

The Korean science, technology and innovation (STI) system does have specific features that are conducive to a MOIP approach. The OECD Reviews of Korean Innovation Policy (OECD, 2023^[1]) provided a detailed overview of the evolution of Korean STI governance, which is aligned with the **three dimensions MOIPs**. **Korea strategically engages** an array of public and private actors in agenda setting, leveraging its history of strategic agendas that direct government support toward specific technologies, sectors, and societal challenges. Advanced tools like foresight, big data analytics, and citizen sentiment analysis are employed to identify priorities, though its agendas have primarily focused on technological priorities. Korea has established multiple institutions to **coordinate** long-term STI priorities, align plans across ministries and agencies, and implement joint cross-ministerial initiatives. This coordination is led by the Presidential Advisory Council on Science and Technology (PACST), which reports directly to the President, while the Ministry of Science and ICT's (MSIT) STI Office supports the process, coordinating STI policies across government. For **policy implementation**, Korea has a comprehensive set of innovation policies covering different sectors and stages of the innovation cycle, with interministerial programmes addressing common challenges. More recently, Korea has adopted challenge-oriented programmes to foster high-risk, high-reward research, and recent updates to STI monitoring and evaluation practices should further enable systemic evaluations that consider the collective impact of various programmes and policies.

Does Korea implement MOIPs?

Korea has launched a series of MOIPs to prioritise technological development in a way that aims to combine traditional economic priority setting with addressing societal goals. This study focuses on three programmes that represent the different types of MOIPs Korea is using: the 3rd Comprehensive Plan to Solve Social Problems Based on Science and Technology (2023-27) (CP3), the National Strategic Technology Policy (NSTP) and the Alchemist. Other MOIPs in the Korean policy mix include the Carbon-Neutral Strategic Technology Policy (CNSTP), the Fine Dust Management Comprehensive Plan, the Future Challenge Technology Development Project, ARPA-H, and the Challenging the Limit Programme.

These programmes have MOIP features that differ from traditional STI governance and policy practices:

- **Strategic orientation** - The NSTP, CNSTP and CP3 are overarching MOIPs, which have collectively developed missions with implicated stakeholders, and then worked across government to develop roadmaps with more details on how they plan to achieve their missions.
- **Policy co-ordination** - Korea's MOIPs are going further than traditional STI co-ordination mechanisms (mainly efficiency-driven and focused on how to avoid duplication) by effectively bringing public and private actors together to identify gaps that need to be addressed to achieve the missions. This co-ordination happens at committees specific to each mission under the CP3 and for each strategic technology with the NSTP, which are comprised of relevant stakeholders from government, industry, and academia.
- **Policy implementation** - These MOIPs mobilise a diverse set of policy levers to achieve the missions, including R&D investments in key areas, skills development, regulatory sandboxes, and procurement for innovation. This wide variety of tools is needed to tackle the complex challenges each mission identifies. The governance and implementation following the programme approval stage for these programmes could go further in ensuring projects and initiatives are more aligned towards the goals through a dedicated portfolio management approach. Some of the MOIPs have less certainty regarding how much funding their missions will receive, while others have national funds earmarked to them over several years and are prioritised during the annual budget process - which are important processes to ensure their success.

Of the MOIPs this paper studied, only the Alchemist's has had time to fully offer its programming over several years. Its challenge-oriented approach has been recognised as a successful programme by Korean authorities and other ministries are seeking to replicate it. The Alchemist is focused on technology development for improving society through a portfolio of research projects. It is more focused on R&D than the other two MOIPs. The latter plan to take more systemic approaches to tackle national missions that would inspire and engage public and private actors across all departments and sectors. It is too early to assess the added value of these other initiatives in the Korean STI policy landscapes.

What could be next steps for MOIPs in Korea?

These policies have added more directionality to the Korean innovation system and are helping to encourage more transformative technological progress and societal change. To help embrace the transformative nature of missions beyond the strong foundation in place, the OECD has identified options for change: **The overall mission-oriented approach** should involve setting a small number of transformative missions for societal challenges with clear, inspiring, and powerful directions at the highest level of government to galvanize national support. To **strategically orient current missions**, Korea should raise the profile of societal missions to encourage greater support and enable a more systemic approach at the national level, while also creating more coordinated priority-setting among its various ARPA-style challenge programmes to complement the overarching mission-oriented framework. For **implementing current missions**, Korea should establish a dedicated portfolio management structure to provide proactive and holistic steering of mission-oriented activities, provide flexible, dedicated, multiyear funding to unlock their transformative potential, and align STI support across government to enhance the likelihood of mission success. Additionally, pathways to market should be developed in current and future challenge-oriented programmes to maximize the impact of developed technologies. Finally, mission monitoring and evaluation processes should be adopted to capture specific systemic effects and assess added value compared to traditional STI policies.

1 Introduction

The work on mission-oriented innovation policy (MOIP) was launched by the OECD Committee for Scientific and Technological Policy (CSTP) in 2019. It aims to provide a comprehensive understanding of how governments try to address societal challenges through various types of MOIPs in different national and thematic contexts, and offers options for change on how to improve these policies based on research of international cases.¹

Using a dedicated methodological framework, this strand of work conducts two main types of fieldwork activities:

1. Systematic investigations: identification and characterisation of MOIP initiatives in all OECD countries and beyond; and
2. In-depth investigations: thorough analysis of mission-oriented policies in a sample of voluntary countries through national reviews of MOIPs.

To date, four countries have engaged in a partnership with the OECD to conduct a MOIP national review: Austria, Japan, Korea and Norway.

The present document sets out the results of the Korean review of MOIPs. The main objectives of this review are to:

- identify and analyse the institutional factors that may foster or impede the design and implementation of MOIPs for societal challenges in Korea.
- systematically scan through the Korean policy landscape to identify and characterise initiatives that match – to a greater or lesser extent – the MOIP’ definition.
- provide a series of options for change to inform future policy decisions regarding Korean MOIPs.

The final goal of the review is to identify the opportunities for and challenges to mission-orientation in Korea, derive corresponding conclusions and options for change, and contribute to discussions on how Korea could, if deemed necessary, strengthen its mission-oriented policy. This should help Korea to more effectively address societal challenges and meet national priorities.

The study was undertaken with financial and substantive support from the Science and Technology Policy Institute (STEPI).

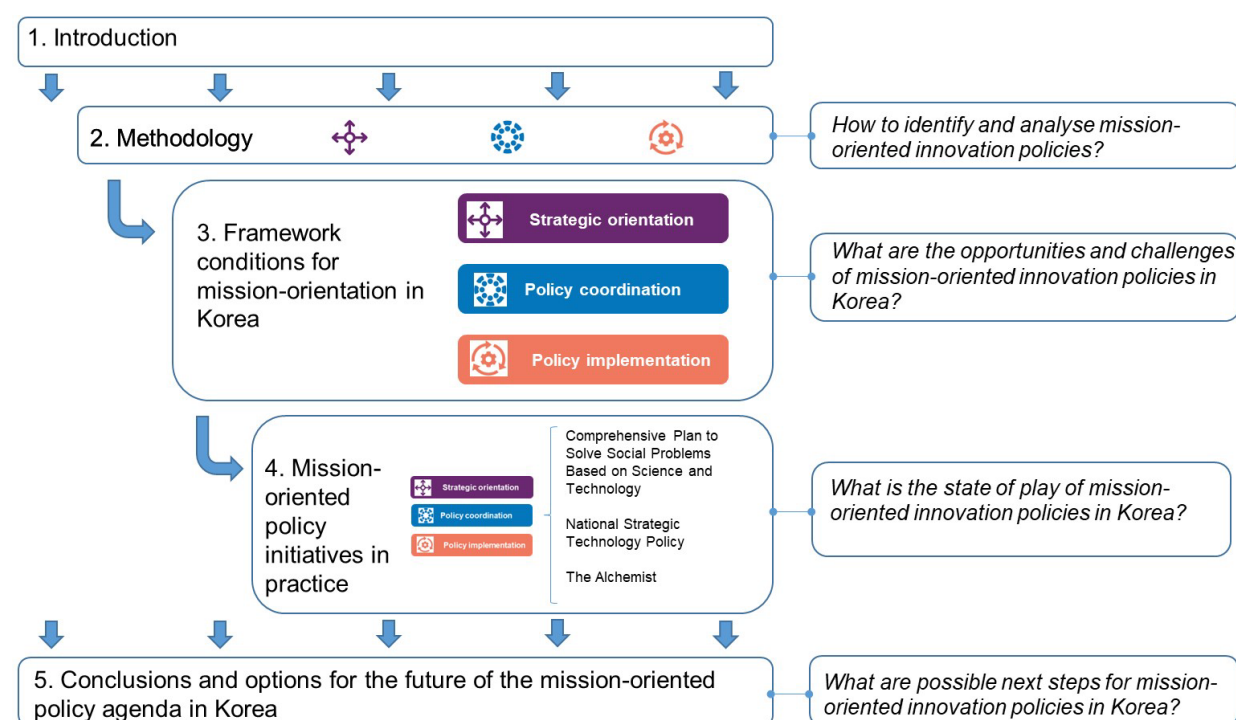
This document has five main sections (Figure 1.1). Following this introduction, the second section synthesises selected elements from the project methodological framework, in particular the definitions of the main concepts used to structure the analysis. The third section (and in greater detail Annex A) examines the MOIP framework conditions (i.e., strategic orientation, policy co-ordination and policy implementation) to address the following questions:

1. How Korea has currently and in the recent past set directions to achieve particular objectives of national importance, in particular those related to societal challenges;

2. How Korea co-ordinates STI policies across policy fields, communities and sectors; and
3. How Korea implements specific STI policies and programmes that match to a greater or lesser extent the features of MOIPs as defined in this study.

The fourth section reviews recent or on-going MOIPs in Korea: the 3rd Comprehensive Plan to Solve Social Problems Based on Science and Technology (2023-27) (CP3),² the National Strategic Technology Policy (NSTP),³ and the Alchemist⁴. A last section recalls the main findings, concludes on opportunities and challenges of mission-orientation in Korea and proposes some options for the next steps.

Figure 1.1. Structure of the mission-oriented innovation policy review



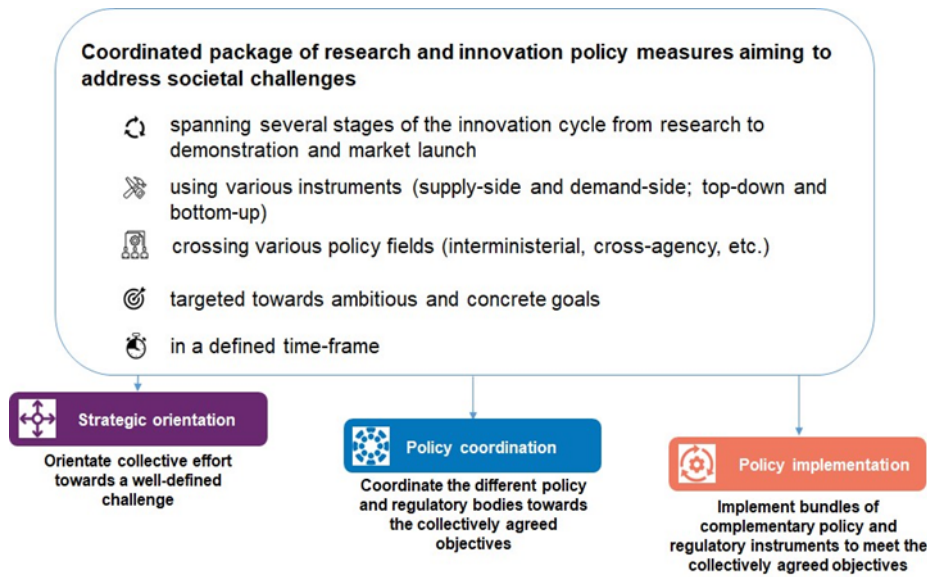
2 Mission-oriented innovation policy review methodology

The methodological framework developed in the context of the CSTP project on MOIPs is used to support and structure the review.

The definition of MOIPs

The definition of MOIPs used in this project builds upon an extensive literature review and is consistent with the main research works conducted in this area (European Commission, 2018^[2]), (European Commission, 2018^[3]), and (Mazzucato, 2018^[4]). A MOIP is defined as a co-ordinated package of policy and regulatory measures tailored specifically to mobilise science, technology and innovation in order to address well-defined objectives related to a societal challenge, in a defined timeframe. These measures possibly span different stages of the innovation cycle from research to demonstration and market deployment, mix supply-push and demand-pull instruments, and cut across various policy fields, sectors and disciplines (Larrue, 2021^[5]).

This definition can be broken down to isolate three main types of functions that MOIPs are expected to fulfil: strategic orientation, policy co-ordination and policy implementation (see Figure 2.1).

Figure 2.1. The definition of MOIPs and the three MOIP dimensions

The three dimensions of mission-orientation

The definition of MOIPs encompasses three main dimensions (Table 2.1). These dimensions were used for fieldwork (design of templates and interview guidelines) and for analysis and delivery of results at national and initiative levels (structure of chapters 3 and 4).

Table 2.1. The three dimensions of mission-orientation

 Strategic orientation	The ability to engage a wide array of public and private actors in the selection of well-informed societal challenge(s) and the legitimization of focused policy interventions towards these challenges.
 Policy coordination	The ability to ensure the consistency of the action frameworks of different public and private bodies covering various policy fields and actors in order to address the targeted societal challenges.
 Policy implementation	The ability to implement, monitor and evaluate in a consistent way an integrated package of instruments / modes of interventions covering different sectors/areas, stages of the innovation cycle and/or disciplines to address the targeted societal challenges.

Types of mission-oriented innovation policies

As detailed in Table 2.2, there are four main types of MOIPs. Among them, ‘Overarching mission-oriented strategic frameworks’ (broader and more top-down initiatives that aim for system-wide transformations) and ‘Challenge-based programmes’ (inspired to a greater or lesser extent by the United States ‘ARPA model’) are the most frequent types of MOIPs currently implemented in countries.

Table 2.2. Basic characteristics of the four main types of MOIP

Type	Leadership	Missions
------	------------	----------

Overarching mission-oriented strategic frameworks - Broad initiatives set up at the highest level of policy making to co-ordinate actions among a wide array of public and private actors toward missions with concrete and ambitious targets.	• Centre of government • High-level committee	• Multiple missions or mission areas • Pursuing ambitious challenges • Long-term horizon
Challenge-based programmes and schemes - Targeted agency schemes, aims to accelerate the development of concrete solutions to a specific challenge	• Agency	• Focused • Seeking acceleration of (most often technological) innovation • Mid- to long-term horizon
Thematic mission-oriented programmes - Traditional thematic research programme reformed to make them progressively more directional, co-ordinated across sectors and disciplines and broader in their means of intervention	• Ministry • Agency	• Focused on competitiveness in the research consortia of the 1980s – 1990s • Mix of societal and competitive challenges in current programmes
Ecosystem-based mission programmes - Mechanisms delegating to ecosystems of actors the development and implementation of strategic agendas to address societal challenges	• Ministry • Agency	• Innovation agenda developed by the innovation actors themselves, with neutral support from public authorities

Source: Larrue, P. (2021^[6]), “The design and implementation of mission-oriented innovation policies: A new systemic policy approach to address societal challenges”, *OECD Science, Technology and Industry Policy Papers*, No. 100, OECD Publishing, Paris, <https://doi.org/10.1787/3f6c76a4-en>.

Korea has a series of mission-oriented innovation programmes, which are discussed in greater detail in Chapter 4. Three cases are highlighted in greater detail: CP3, the NSTP, and the Alchemist. These programmes were selected because they provide a representative snapshot of the types of MOIPs Korea has deployed. The first two are variations of overarching mission-oriented strategic frameworks, but the CP3 is more societal focused and is the third iteration (becoming mission-oriented in this latest plan), and the NSTP is more technical and the first iteration. The Alchemist is a challenge-based programme, and its early signs of success have spurred a series of similar initiatives across the Korean government. These variations in programme type allow for a more varied analysis in the study.

3 Framework conditions for mission-orientation in Korea

The MOIP framework conditions systematically define the national factors that influence the need for and ability of a country to design, fund and implement MOIPs. This chapter will use the three dimensions that lie at the core of MOIPs (strategic orientation, policy co-ordination and policy implementation) to explore how conducive Korea's existing STI system is to pursue a mission-oriented approach. The main underpinning hypothesis is that the forms and types of MOIPs in a given country can only be understood within:

- the national institutional setting in which they are embedded and that largely determines what is politically, administratively, and socially feasible at one point in time. Countries differ for instance in their capabilities to steer, fund and implement complex policies (such as MOIPs); and
- the policy trajectory that has gradually formed them, moving forward through experimentation, negotiation, and learning. These trajectories are driven by several factors, including the general policy stance toward state intervention, the respective choice of policy instruments (e.g., generic vs. specific measures).

This analysis of the national institutional setting and how it developed overtime builds on the work from the OECD Reviews of Innovation Policy: Korea 2023 (OECD, 2023^[1]). Annex A provides a synthesis of the evolution of the Korean STI governance system, which is discussed in more detail in the Review. Table 3.1 summarises how Korea's approach to STI policymaking is already strongly aligned with the three dimensions of mission-orientation (Table 3.1).

Table 3.1. The three dimensions of mission-orientation

	Characteristic definition	Existing Korean approach
 Strategic orientation	The ability to engage a wide array of public and private actors in the selection of well-informed societal challenge(s) and the legitimization of focused policy interventions towards these challenges.	Engages a wide array of public and private actors in agenda setting. Has a history of using strategic agendas to provide direction and channel government support to certain technologies and sectors.
 Policy coordination	The ability to ensure the consistency of the action frameworks of different public and private bodies covering various policy fields and actors in order to address the targeted societal challenges.	Brings together a variety of public- and private-sector actors to co-ordinate on STI policy through PACST. Provides the STI Office with a strong mandate and powers to centrally co-ordinate STI policies.
 Policy implementation	The ability to implement, monitor and evaluate in a consistent way an integrated package of instruments / modes of interventions covering different sectors/areas, stages of the	Has a well-established set of innovation policies to cover different sectors/areas, and stages of the innovation cycle

innovation cycle and/or disciplines to address the targeted societal challenges.
--

Strategic orientation

Korea's STI policymaking process seems to lend itself well to a mission-oriented approach. Korea uses central agendas for identifying both technological and sectoral priorities, an approach that has mission-oriented characteristics. This is epitomised by a series of nested strategies that guide Korean STI policy: a 10-year long-term vision, the presidential programme agenda, the five-year S&T Basic Plan, the mid-to long-term National R&D Investment Strategy, and annual plans for each ministry. Ad-hoc strategies (e.g., focused on technologies) complement this work. Korea embeds many of these strategies in legislation that requires them to be updated on regular cycles, which helps continue this work beyond political election cycles. Similar legislation has been used to ground the mission-oriented policies discussed in Chapter 4 (e.g., the CP3 and NSTP, which both must be updated every five years).

Box 3.1. Korean green strategies

Korea has had two overarching green strategies with mission-oriented qualities: the Green Growth Strategy⁵ (2008) and the 2050 Carbon Neutrality Strategy⁶ (2020) (Eum, 2023^[7]).

The 2008 strategy set out the government's vision to become one of the top 7 green superpowers in the world by 2020 and amongst the top 5 by 2050. This work was underpinned by the Framework Act on Low-Carbon Green Growth⁷ (2010). An extensive governance system oversaw cross-government implementation, headed by the Presidential Committee for Green Growth and the Green Growth Planning Office, and then 'Chief Green Officers' were created within each ministry to facilitate communication between ministries and committees. The Green Growth Strategy was accompanied by three directions, with 10 specific tasks that were largely STI focused (Green Growth Committee, 2009^[8]).

In 2020, Korea's 2050 Carbon Neutrality Strategy set out a vision for how it will achieve its Paris Agreement targets as part of its broader COVID-recovery work. It focused on five areas, and seven sectors. It was overseen by Ministerial Meetings on the Korean New Deal, chaired by the Deputy Prime Minister of Economy, and supported by a working-level taskforce under the Minister of Economy and Finance (Eum, 2023^[7]). It included a broader range of actions than the 2008 strategy, including policy innovation (e.g., including climate impact in budgetary decisions and establishing a carbon price system), societal change (e.g., raising public awareness and expanding public participation in decision making processes), and technological advancement (e.g., improving STI co-ordination across government, and investing in more research around core green technologies) (Ministry of Environment, 2020^[9]).

Following elections in 2022 and a change in political parties, Korea's work towards reaching carbon neutrality was shifted towards a new framing and governance model. Notably, a new set of technology-focused missions were established through the Carbon-Neutral Strategic Technology Policy that are a flagship policy of this government (see Box 4.5).

While Korea's STI strategies have historically helped drive Korean progress by focusing on certain technology areas, its new mission-oriented plans represent a step forward by including a broader set of

policy actors to achieve more transformative objectives. Korea also has an almost 20-year history of identifying societal challenges that it seeks to address using STI policy (i.e., the previous iterations of the Comprehensive Plan to Solve Social Problems Based on Science and Technology). These previous plans did not have concrete goals and governance frameworks to address priority issues, which resulted in more fragmented implementation than today. Korea's new mission-oriented approach is providing actors more concrete direction, and accompanying governance to encourage a more systemic approach across government.

Korea has well established processes to inform its strategies using foresight and stakeholder consultations. The government's combination of bottom-up broad consultation to inform its strategies, is coupled with top-down decision making on the ultimate orientation. This is similar to what (Larrue, 2021^[5]) describes in the context of missions, and how "the orientation process [for missions] unfolds as a succession of top-down and bottom-up phases, opening and reducing the space of potential options through a mix of concertation and selection stages."

At the same time, the Korean government has found its traditional sectoral strategies do not provide enough directionality to the innovation system for pursuing particularly breakthrough research. In a 2020 report on mission-oriented high-risk R&D, it lamented the fact that its competitors like the United States and Japan have successfully spurred disruptive innovations that have brought about new sources of competitiveness through ARPA-style challenge programmes. The same report outlines how Korea produces a lack of useful patents that lead to revenue generation, "and that more than 90% of R&D is focused on short-term results". The report suggests that Korea should strive to shift from being a "fast follower" to a "bold and challenging first mover". This requires a R&D support system that promotes mission-oriented high-risk R&D that could lead to more disruptive innovations (KISTEP, 2020^[10]).

Korea is now expanding its focus on using STI to tackle social problems through a broader mission-oriented approach. The government's commitment to these types of policies is highlighted in the most recent iterations of its major STI planning documents for directing STI priorities and programme development:

- The current long-term vision from 2020 (*Innovate KOREA 2045 – Challenges and Changes for the Future*⁸) says the country must establish clearly-defined missions to better organise ministries and agencies, and their work in tackling broad social challenges (MSIT, 2020^[11]).
- Of the 120 tasks under the 2022 presidential programme agenda, four involve mission-oriented policies, including establishing a mission-oriented approach in responding to national challenges (e.g., carbon neutrality and aging). Another task mandates the Ministry of Science and ICT (MSIT)⁹ to take a whole-of-government approach with a mission programme that leverages investment from government, industry and academia (President of the Republic of Korea, n.d.^[12]).
- The 5th Science and Technology Basic Plan 2023-27¹⁰ identifies best practices and policy trends from other countries – several of which are mission-oriented innovation policies. It then identifies five main tasks for the next five years, the first of which is to develop a mission-oriented R&D strategy that encourages industrial development and addresses social issues. It goes into detail on how this should be delivered, including setting time-bound goals, establishing roadmaps, creating 'vertical' budgets for each ministry and 'horizontal' budgets for the missions, and creating a cross-ministerial approach for R&D planning, investment, and evaluation. It says a portfolio approach should be used, which could include standards, regulations, R&D investment, and support for commercialisation. The first task it identifies is the National Strategic Technology Program Basic Plan (2024-2028), which outlines the NSTP's mission-oriented approach for encouraging technology leadership (MSIT, 2022^[13]). The same priorities are echoed in the National Research and Development Mid- to Long-Term Investment Strategy (2023-2027), which identifies

the NSTP and Carbon-Neutral Strategic Technology Policy (CNSTP) as the two major actions for delivering on a mission-oriented approach (MSIT, 2023^[14]).

Box 3.2. France's Acceleration Strategies – an example of horizontal budgeting

France's Acceleration Strategies (SAs) were created as the main new tool for the implementation of the Investments for the Future programme 4 (PIA4). In contrast with the former PIAs, SAs were created to add 'directionality' and a 'systemic perspective' to the PIA. Since 2021, the PIA has been incorporated into the France 2030 plan. The plan has ten overarching objectives which encompass 24 SAs. Each SA is driven by a dedicated strategy that was developed with significant involvement of relevant stakeholders and experts, with inputs from public consultations. The SAs have a dedicated central coordinator – most of them are located within the General Secretariat for Investment (SGPI), i.e. the administration in charge of the execution of France 2030 under the Prime Minister's Office.

The Acceleration Strategies support a wide range of activities from research and innovation to, for instance, training and market legitimacy. The selection and funding of activities and projects are managed by four agencies. The type of support and instruments they mobilise range from research subsidies, institutional funding to intermediary organisations and equity investment in business companies via seed and growth funds, or funds of funds. Some of them include scale up and deployment mechanisms.

The PIA is funded by extra-budgetary resources, which are in great part directly allocated to funding agencies, without influence by their supervisory ministries. This approach provides more centralised steering (since the investment decisions are taken by the SGPI under authority of the Prime Minister) and more flexibility in funding allocation and management (as they are not subject to normal state budget regulations). While during the PIA1 to PIA3 the budget were 'sectorised', i.e. allocated to broader actions (higher education, technology transfer, etc.) or sectors (transport, energy, defence, etc.), they are in the PIA4 integrated for each Acceleration Strategy. This means that the different actions or sectors will be mobilised and staged as needed, according to the progress of implementation of each of the strategy.

Source: OECD Mission-Oriented Innovation Policy Online Toolkit, <https://stip.oecd.org/moip>.

These strategic documents provide the policy rationale and legitimacy for Korea's MOIPs, as they are directly cited in the CP3, NSTP and CNSTP. These MOIPs have then collectively developed missions with implicated stakeholders and worked across government to develop roadmaps with more details on how they plan to achieve their missions (see Chapter 4 for more information).

As in many countries, Korea's innovation policy is somewhat shared between the minister of science (MSIT) and industry (the Ministry of Trade, Industry and Energy (MOTIE)¹¹). While MSIT supports MOTIE's prioritisation processes, the two ministries could co-ordinate more closely when setting priorities. They each have released strategies that identify different sectors and technologies, and their respective programmes provide preferential supports for different areas. MOTIE has identified 12 fields and 71 national core technologies (MOTIE, 2021^[15]), and 4 high-tech strategic technology fields and 17 detailed technologies to focus its applied R&D supports towards (MOTIE, 2022^[16]). MSIT has identified 12 strategic technologies with 50 sub-technologies under the NSTP, and 17 strategic technologies with 100 sub-technologies under the CNSTP. The Ministry of Economy and Finance (MOEF)¹² also has preferential support it offers through R&D tax credits aimed at 'new growth and key enabling technologies' (235

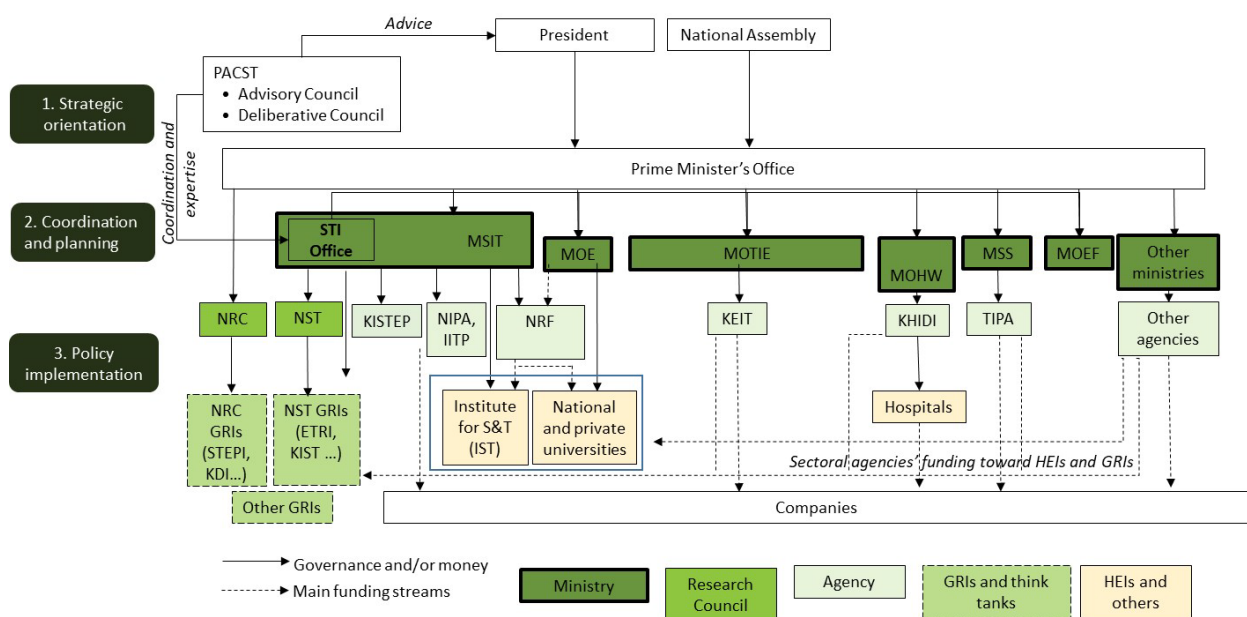
technologies in 12 areas) and seven national strategy technology areas (some of the NSTP technologies but not all of them). These technology priorities and missions are in addition to the technology challenges under the Alchemist (which alone has 20), and those under development through the Future Challenge Technology Development Project, ARPA-H, and the Challenging the Limit Programme. Furthermore, the CP3 has identified 4 missions for societal challenges and has more in development.

Through its MOIPs, Korea is providing more direction to the STI system, particularly in encouraging R&D to drive technological leadership in priority areas and, to a lesser extent, to address societal challenges. Through a set of new strategies, it is taking a more systemic approach to STI policymaking.

Policy co-ordination

Along with its history of agenda setting, Korea has an existing suite of co-ordination mechanisms in the STI field that lend themselves well to mission-oriented approaches. Figure 3.1 presents the Korean STI governance structure across the ministries and agencies that support R&D in Korea. A more complete picture would include a far greater number of ministries and agencies that develop policies and manage R&D programmes.

Figure 3.1. Korea's STI governance structure



Note: Note: Full names of institutions here in descending order: Presidential Advisory Council on Science and Technology (PACST), Ministry of Science and ICT (MSIT), Ministry of Education (MOE), Ministry of Trade, Industry and Energy (MOTIE), Ministry of Health and Welfare (MOHW), Ministry of SMEs and Start-ups (MSS), Ministry of Oceans and Fisheries (MOF), National Research Council for Economics, Humanities and Social Sciences (NRC), National Research Council for Science and Technology (NST), National IT Industry and Promotion Agency (NIPA), Institute of Information and Communications Technology Planning and Evaluation (IITP), National Research Foundation (NRF), Korea Evaluation Institute of Industrial Technology (KEIT), Korea Health Industry Development Institute (KHIDI), Korea Technology and Information Promotion Agency for SMEs (TIPA), Korea Institute of S&T Evaluation and Planning (KISTEP), Korea Institute for Industrial Economics and Trade (KIET).

Source: OECD (2023^[1]), *OECD Reviews of Innovation Policy: Korea 2023*, OECD Reviews of Innovation Policy, OECD Publishing, Paris, <https://doi.org/10.1787/bdcf9685-en>.

Two notable organisations that provide overarching direction and oversight to Korea's STI system are:

1. The Presidential Advisory Council on Science and Technology (PACST)¹³, currently chaired by the President, was created in 1991 to provide strategic, medium- to long-term advice on S&T, and co-ordinates R&D policy across the ministries (except defence, which is treated separately from the rest of the system). It focuses on national tasks that require comprehensive co-ordination across ministries and relevant stakeholders, and that require a decision from the highest level of the S&T community, namely, the President. The PACST has an advisory body, which provides high-level strategic advice to the President, and a deliberative body, which focuses on planning, budget formation and monitoring of R&D programmes.
2. The Science, Technology and Innovation Office (STI Office) supports MSIT's horizontal co-ordination function across the entire government. The STI Office plays the role that is often referred to as the "STI control tower" and has a powerful mandate, defined by law. It leads key processes to set orientations, co-ordinate plans and budgets, and monitors and evaluates policy implementation. It is staffed with specific subject matter experts and has an extensive set of more specific responsibilities. Sectoral ministries must submit their annual R&D action plans, and mid-term action plans to the STI Office for review.

The budget process is an important part for how both PACST and the STI Office provide co-ordination across the Korean STI system. The STI Office sets R&D investment directions that the other ministries use to guide their budget requests, and then both the STI Office and PACST review proposed R&D programmes before the MOEF makes the final decision. This oversight from the STI office ensures sufficient funding is allocated to certain priority areas, and prevents duplication from across government (see Annex A for more details on this process).

The established governance co-ordination mechanisms through PACST and the STI Office support Korea's MOIPs. PACST has played a major role in providing policy guidance and approvals of key elements under the new mission-oriented initiatives. The PACST Deliberative Council produced a guidance document in October 2022 that outlines how to take a mission-oriented approach, including how to design missions, take a more flexible approach to investing, and evaluate their performance (PACST, 2022^[17]). The role of PACST in the individual initiatives is explained more each one in Chapter 4, but its general role has been to approve the approaches for each MOIP, and the thematic priority areas. The STI Office also plays an important role in the co-ordination of MOIPs. It has led the design of several MOIP overarching policies (e.g., the NSTP) and individual initiatives (e.g., the Innovation Challenge Programme in 2018-2020).

The budget process plays an influential role in supporting MOIPs. As part of the STI Office's review process of budget proposals, amongst other factors, it makes decisions based on how the proposals are linked to government priorities – including the NSTP, CNSTP and CP3. The NSTP and CNSTP have notional budgetary targets under the 5-year R&D Investment Strategy that provides them with more certainty that they will receive funding for their mission areas than the CP3. This certainty is further reinforced by the STI Office's review, which in recent budgets has particularly prioritised proposals related to the NSTP and CNSTP. The actual budget for these policies is determined by tallying the successful commitments at the end of each budget cycle.

The Korean STI system underwent major reforms in 2023, with budget cuts for some areas of the STI system, and an explicit emphasis on supporting mission-oriented innovation policies (notably technologies aligned with the NSTP and CNSTP) and international co-operation. As a result, in the 2024 National Research and Development Project budget allocation/adjustment, 5 trillion won (~3.4 billion euros) was allocated to carry out work related to National Strategic Technologies, which represented a 7.4% increase

from 2023 (MSIT, 2024^[18]). At the same time, proposals related to the mission-oriented CP3 did not receive the same level of priority, so the 2023 reformers resulted in cuts for the budget allocated to social problem-solving R&D.

Each MOIP has its own governance. High-level committees are responsible for each initiative and encourage synergies between missions. These committees often sit under PACST and seek its approval at various stages. The individual missions then have their own governance committees, which develop roadmaps, often with the help of the STI Office, for how they will address their missions and what interventions will be needed from across government (explored more in Chapter 4).

Korea has a well-established co-ordination system that brings congruency to its STI policies across government. This system is encouraging sectoral ministries to commit to overarching missions. It is reinforced by additional mission-level governance.

Policy implementation

Korea's STI policies are mainly implemented in the form of R&D programmes by around 15-17 dedicated R&D agencies, which act on behalf of ministries. Agencies can be mandated to allocate funding and manage projects (R&D management agencies), while others perform a range of other actions to support R&D and innovation (R&D promotion agencies). In some cases, the difference between the agency roles is unclear, with some agencies playing both.

An important aspect of Korean innovation policy is the conditionality it places on its funding. Technologies that are commercialised from government-funded R&D projects are charged technology fees. This helps "maintain a virtuous cycle of technology development" by reinvesting the fees in further research and development through a quasi-dividend arrangement (Related Joint Ministers, 2024^[19]).

Korea has the second-highest R&D intensity among OECD countries; its gross domestic expenditure on R&D (GERD) represented 5.21% of GDP in 2022 (OECD, 2024^[20]). It is also among the OECD countries that provide the largest level of total government support to business R&D (BERD) as a percentage of GDP, at a rate of 0.35% of GDP in 2021 (OECD, 2024^[21]). The allocation of public funds is rooted in the historical context of a country, including the pace of development, industrial structure, and government policies. A large portion of Korea's government appropriations dedicated to R&D is thematically oriented (76% of the total (direct) funding for R&D in 2022, in comparison to 84% in the United States, 65% in Japan, and 48% in Germany). The country dedicates a relatively small proportion of its government R&D budget to projects and activities related to health and society issues due to its historical need to invest more in defence. In 2021, 14% of R&D funds were allocated to health and society, compared to 28% in Norway, 25% in the United States and 22% in Germany (OECD, 2024^[20]).

Korean policies to provide more directionality to innovators, particularly in using STI policy in addressing social issues, have evolved over almost two decades – moving from broad priority areas to more time-bound targets with accompanying roadmaps. One important mechanism to deliver on these cross-cutting priorities has been inter-ministerial programmes that combine resources and capabilities from across government to tackle common challenges.

In addition, Korea is growing the number of challenge-based programmes it offers. In 2014, the government released Competitive R&D Promotion Guidelines to establish models for different types of competitive R&D (summarised in Table 3.2).

Table 3.2. Korean models for competitive R&D programmes

Category	Description	Suitable businesses
----------	-------------	---------------------

Tournaments	Through step-by-step mid-term evaluation for a specific project, some are eliminated, and some projects are continuously supported	Large business Mid to long term business
Competition planning	At the project planning stage, two to four times as many research institutes are selected to conduct planned research and the planning results are evaluated to select an R&D performing institute.	Large business Mid to long term business
Postpaid type survival	After multiple research groups perform the same task, research funds are paid differentially by evaluating the excellence of the final results.	Small business Short term business
Parallel type	Different teams are selected to pursue a different approach to tackle the same research goal. There is a mid-term evaluation, at which point some tasks selected for further support	Small to medium sized business Short to medium term business

Source: KISTEP (2021^[22]), 2021 년 과학기술혁신정책지원사업 도전적·혁신적 R&D 촉진을 위한 법제 개선 연구 [2021 Science and Technology Innovation Policy Support - Project Research on improving legislation to promote challenging and innovative R&D], https://www.kistep.re.kr/reportDetail.es?mid=a10305030000&rpt_tp=831-002&rpt_no=RES0220220037.

The R&D Innovation Plan (2018) announced a strategy focused on using a challenge-oriented approach to pursue high-risk, high-reward research, which set the foundation for more institutionalised approaches for challenges and “mission-oriented problem-solving R&D”. Following the Basic Plan, challenge-oriented R&D projects were included in the Framework Act on Science and Technology, and the 2018 National R&D Innovation Plan (KISTEP, 2020^[23]).

Korea’s model of challenge-based programmes is now coalescing around a similar method: programmes set a goal, and then programme managers oversee a suite of projects and stage-gate the funding to eventually double-down on one approach that seems most promising. After experimenting with several different challenge-based programmes (keeping some and phasing others out), Korea plans to introduce at least three more programmes based on this model (see section 4.3 for more information).

While this approach to supporting R&D is more directional, it may add more complexity to the Korean system by adding too many priorities. With more challenge-based programmes being created, there is an opportunity to align their priorities with the existing roadmaps from overarching MOIPS rather than creating new priorities.

4 Mission-oriented policy initiatives in practice

This section presents three mission-oriented case studies. See Annex B for a summary of each case, presented across the three MOIP orientations (i.e., strategic orientation, policy co-ordination and policy implementation).

The Comprehensive Plan to Solve Social Problems Based on Science and Technology

The 3rd Comprehensive Plan to Solve Social Problems Based on Science and Technology (2023-27) (CP3) is an overarching mission-oriented innovation policy that aims to realise a happy society by solving social problems that matter most to Koreans through science and technology.

It is rooted in the Presidential Advisory Council on Science and Technology's 2006 report Technology-based Quality of Life Enhancement Plan for Vision 2030, which proposed: (1) diagnosing the current situation; (2) linking the expansion of supply of technology to demand; (3) developing technology and building infrastructure to improve Korea's quality of life; (4) establishing technology support system for the socially disadvantaged; and (5) preparing comprehensive policy support plans (STI Office, 2006^[24]). A first attempt at this approach was the 2007 initiative called Comprehensive Measures to Improve the Quality of Life Based on Technology, which, along with the first two iterations of the Comprehensive Plan to Solve Social Problems Based on Science and Technology, encouraged ministries across government to prioritise R&D projects that address specific social problems and sub-priorities for each problem.

This progression in Korea's policy approach reflects a broader pattern seen in mission-oriented innovation policies globally. The degree to which policies adhere to the three dimensions of mission orientation (i.e., strategic orientation, policy co-ordination and policy implementation) often grows over time for mission-oriented innovation policies around the world. This gradual evolution in Korea from one Comprehensive Plan to the next follows this pattern, as it has gained more concrete goals (now in the form of missions), formal governance, and tools to deliver on its ambitions of addressing social problems. The five-year S&T Basic Plan (2023-27) highlights the CP3 as a tool for encouraging public participation and public-private cooperation to tackle core social problems (MSIT, 2022^[13]). The National Research and Development Mid-to Long-Term Investment Strategy (2023-2027) also mentions addressing social problems, but does not mention all of the CP3's mission areas and prioritises social problems the CP3 has not set missions for (e.g., disaster safety) (MSIT, 2023^[14]). While it has not received the same prominence as other mission-oriented policies (e.g., the NSTP and CNSTP) in these core documents, including the CP3 in these documents demonstrates it is still a priority and that Korea is dedicated to delivering on it and addressing social problems.

The 3rd Comprehensive Plan has two aspects:

1. 10 social problem domains, and 43 social problems under them, for which to focus government R&D investments upon. This approach is similar to how the previous iterations defined social problems – although each plan has altered the list of domains and problems. To address these 43 social problems, Korea has identified 10 broad tasks; and
2. Of these 43 social problems, five ‘core problems’ are being treated as missions, with a set of mission statements, more specific sub priority areas (about 2 per problem), and targets for each of those sub priorities. To date, two of these core problems have been fleshed out in this manner, and the remaining three are under development.

While funding for the CP3 decreased from 2023 to 2024, government investment to address the social problems had been growing – reaching 1.78 trillion won (~1.2 billion euros) in 2022 (6% of the government R&D budget), up from 1.23 trillion (~800 million euros) in 2018. This funding signals the government’s continued commitment to funding solutions related to the CP3.

Strategic orientation

The CP3 provides direction by identifying 43 social problems and five core social problems that it aims to address using STI.

To identify these problems, the government conducted an extensive consultation process involving public hearings, experts, relevant ministries, and public-private advisory councils for social problem solving. The process was led by the STI Office. In 2021, it surveyed industry and academic experts regarding their social priorities, as well as a separate survey of 1,000 members of the general public.

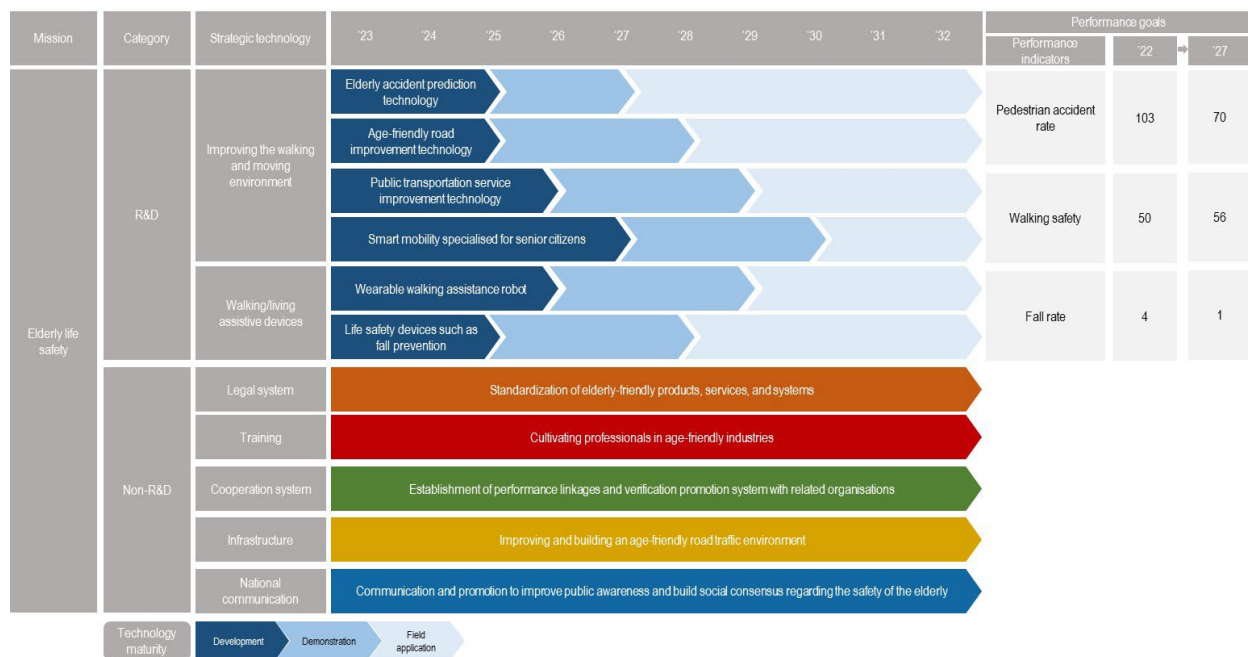
The government conducted big data analysis, where it looked at public sentiment towards key social problems over recent years. This work was coupled with topic modelling analysis that compared news from 2018-19 to 2020-21 to see if any new social problems had emerged, and if the sentiment to the problems under the 2nd Comprehensive Plan had changed in any way. Analysis was then conducted of future social problems through the 6th Science and Technology Forecast Survey – a SWOT analysis from the Korea Institute of S&T Evaluation and Planning (KISTEP) evaluates key social problems that STI has a role in addressing. KISTEP also looked at the results from other foresight studies in Europe and the United States.

These surveys and research were used to propose a new set of social problems, building on the 2nd Comprehensive Plan. MSIT and KISTEP held open discussions with related ministries, agencies, and local governments to collect their opinions on the draft Comprehensive Plan and these new priorities. The final set of 10 problem domains and 43 social problems was preliminarily approved by the Public-Private Council for Social Problem Solving before receiving final approval by the PACST Deliberative Council (see Annex C for more information, including which problems changed or are new for the CP3 and the full list of social problems).

After deciding on the problem domains and social problems, MSIT and KISTEP identified 10 social problems that were candidates to be designated as ‘core social problems’. Another survey was conducted, and the government conducted internal analysis regarding which of these problems were most appropriate for a technological response and necessitated government support. Following this process, five core social problems were ultimately selected for additional support through a whole-of-government mission-oriented approach: ageing, cybercrime, reducing fine particle matter (a long-standing issue for Korea that is explored more in Annex E), reducing microplastics, and reducing household waste (see Annex D for more information on the core problems).

While the other 38 social problems are outlined at a high-level, the five core social problems are framed in a manner that provides innovators with more direction (although because the strategy is relatively new at the time of producing this report, so far only two core problems (ageing and cybercrime) have been fully fleshed out in this manner). For example, one core social problem is ageing. It is then broken down into two target areas – elderly life safety and bridging the digital divide in this case – and each of those has a series of timebound indicators – such as reducing pedestrian deaths and injuries per 100,000 elderly people from 103 in 2022 to 70 in 2027 and reducing the percentage of seniors who report a fall from 4% in 2022 to 1% in 2027. The mission areas are accompanied by a roadmap of R&D and non-R&D tasks that will help achieve their targets (see Figure 4.1 as an example).

Figure 4.1. Elderly life safety roadmap



Note: Pedestrian accident rate (number of people): Number of pedestrian traffic accident deaths and injuries per 100,000 elderly people; Walk safety (%): Elderly people's safety satisfaction with the width of the walking area, obstacle removal and safe road surface treatment; Fall rate (%) The percentage of elderly people who reported that they had fallen

Source: PACST (2023^[25]), 제 3 차 과학기술 기반 사회문제해결 종합계획 (2023-27) (안) [3rd Science and Technology-Based Comprehensive Solution Plan for Social Issues (2023-27) (Draft)], https://www.pacst.go.kr/jsp/council/councilArchiveView.jsp?archive_id=1070&cpage=3.

The government plans to annually establish implementation plans for the CP3, which may include updates on the implementation of the five core problems. Additionally, as societal trends evolve, the government may revise the list of 43 social problems identified under the 10 domains. Each ministry may undergo a performance evaluation, ahead of the next annual implementation plan of the CP3 – although this is not legally mandated.

Policy co-ordination

The Public-Private Council for Social Problem Solving, which is currently under the STI Office, is planned to advise and coordinate on the CP3. According to the Plan, the Council is intended to be moved under

the PACST, specifically under the Steering Committee of its Deliberative Council. There is a secretariat within the STI Office in MSIT that works with the S&T Policy Centre for Social Problems in KISTEP to manage the implementation of the CP3.

In addition to addressing the social problems themselves, the CP3 aims to expand multi-ministry co-operation. It does this through Flagship Programme Planning Committees for each mission. Each of the missions has a Flagship Programme Planning Committee to coordinate R&D programmes to address the mission targets. These committees are chaired by a champion ministry (e.g., the Ministry of Health and Welfare is the lead ministry for bridging the digital divide for the elderly), and their memberships include relevant ministries and agencies, a representative from the Public-Private Council, and technical and non-technical experts (including from government research institutes). MSIT works with these committees to support the planning of inter-ministerial R&D programmes that align with the roadmap for each mission. Additionally, the committees serve as consultative bodies to coordinate matters among participating ministries. It has been more difficult to have ministries commit to supporting the missions through non-R&D measures to date, but these measures are part of the roadmaps and will be more important as the missions mature.

MSIT co-ordinates the funding of R&D for the CP3 like it does for every other R&D program. Korea does not make multi-year budgeting decisions but goes through an annual process described in section 3.2 and in Annex A in more detail. As part of the broader budgeting process, sectoral ministries submit their R&D-related proposals to the STI Office for review which, amongst other factors, makes funding recommendations based on how the proposals are linked to government priorities – including the CP3. Sectoral ministries develop their proposals based on their own strategic priorities, including the commitments they make through the CP3's Flagship Programme Planning Committees. The budget for the CP3 decreased from 2023 to 2024, but government investment to address the social problems had been growing since 2018. While the implementation plans for the CP3 do not explicitly identify connections to the NSTP, many social problems have the potential to utilise and benefit from the technologies being fostered as part of the NSTP (e.g., social problems such as chronic diseases, rare and intractable diseases, mental illnesses, and infectious diseases may benefit from advanced bio technologies, a National Strategic Technology, and sub-technologies including synthetic biology, gene and cell therapies, and digital health data analysis and utilization).

The Public-Private Council for Social Problem Solving also allows local governments to participate, who can make proposals aimed at addressing their needs. To enable this participation, the Plan proposes to work with 10 existing regional problem-solving platforms – which are private sector-led with local government involvement. These platforms aim to identify and solve regional social problems.

In addition, 'hub institutions' (a university or a government-funded research institute) will be identified to accumulate multidisciplinary research results from government-funded research for each of the core problems. These hubs will then disseminate the research and encourage co-operation by government, local government, researchers, companies, and citizens.

The Plan also emphasises the importance of international co-operation for solving some social problems (e.g., transboundary environmental pollutants and radioactive contamination). As a result, it will expand international joint research and development projects, and increase bilateral or multilateral co-operation.

Policy implementation

For the 43 social problems, the strategy has identified 10 tasks across three pillars:

- Strengthening strategy for problem solving
 - (Task 1) Streamlining the governance structure for social problem-solving
 - (Task 2) Identifying major social issues and upgrading the management system
 - (Task 3) Establishing a foundation for strategic investment in social problem-solving R&D
- Strengthening mission-oriented problem-solving capabilities
 - (Task 4) Implementation of mission-oriented problem-solving programs
 - (Task 5) Strengthening user-driven R&D for social problem solving
 - (Task 6) Advancement of social problem solving R&D performance management system
- Creating tangible results led by users and the private sector
 - (Task 7) Strengthening the capabilities of social problem-solving entities
 - (Task 8) Stimulating the ecosystem of collaboration and sharing for social problem-solving
 - (Task 9) Establishment of a system to deliver R&D performance to problem sites
 - (Task 10) Strengthening private sector-led dissemination of excellent outcomes to the field

The core problems' roadmaps have identified R&D and non-R&D priorities. For the R&D priorities, these include specific areas for technological advancement, and the timeframe they hope to see these technologies evolve from development to deployment. For the non-R&D priorities, the roadmaps include technology diffusion, training, infrastructure, public communication, and setting technology standards.

The S&T Policy Centre for Social Problems in KISTEP identifies and analyses issues related to the social problems outlined in the plan, as well as emerging ones. New inter-ministerial programmes are developed through the Flagship Programme Planning Committees. They are informed by inter-ministerial working-level discussions, public forums with citizens and experts, and hackathons with stakeholders. For particularly urgent or complex issues, they may be escalated to the PACST for high-level discussion and decision-making. Cross-cutting issues are brought to the Public-Private Council to initiate co-operation between relevant ministries and agencies.

Successfully developed technologies will receive follow-on support to encourage commercialisation. This includes the linkages with public procurement, follow-up funding in the next year's budget, and access to regulatory sandboxes (see Box 4.2 for more information on the use of procurement for innovation). Some municipalities operate 'Pre-Sandbox Programs' that support research on regulations before applying for a regulatory sandbox.

Box 4.1. Procurement for innovation in Korea

Korea has several innovative measures to help the government identify and procure Korean innovations. For instance, products can be listed on the Public Innovation Procurement Platform, which is an open marketplace that promotes products to government departments. There are several pathways to be listed on this platform. One way is for a sectoral ministry to nominate 'Excellent R&D Products' (if they are the result of publicly-funded R&D) or 'Prototype Product' (if they are pre-commercialisation). They must meet a set of other standards set by the ministries and be confirmed by MOEF's Public Procurement Policy Council. The second pathway allows either innovators (potential suppliers) or government customers (public organisations who want to purchase a product) to nominate products, and then the evaluation is conducted by the MOEF's Public Procurement Service (PPS), and confirmed by the Public Procurement Policy Council.

The 'Innovative Product Scouter' Programme also helps innovative products enter the government procurement market by discovering products that are struggling to enter the procurement market or are outside the existing procurement market, and then providing them with targeted support.

These initiatives are both used as part of the toolkit for the CP3 to help deploy solutions. Under the Comprehensive Plan, there will also be an exemption of the 'technology fees' (see section 3.3 for a description of these), which the government would otherwise charge on revenues from the commercialisation of government-funded research, including when purchasing products produced by companies utilising public technology.

Citizen engagement is also a major focus for the Plan. It will expand localised problem solving and citizen science by utilising regional universities and local research institutions as regional problem-solving platforms, and promoting co-operation with residents, researchers, and local governments to resolve local issues. It will communicate new technologies to help build social acceptance around them and identify actions individuals can take through the Social Problem-Solving Platform (<https://www.ntis.go.kr/scisopplatform/main.do>), an online resource managed and updated by KISTEP that provides information on social problems and related R&D efforts. The Plan will also partner with organisations dedicated to science and culture to further promote citizen science and engagement (National Science Museum, Korea Foundation for the Advancement Science & Creativity (KOFAC), etc.).

The CP3 has two main categories of indicators:

1. It has targets to have 65% of the population perceive STI funding contributes towards addressing the social problems, and 40% of the budget for social problem-solving R&D be for applied social problem-solving R&D by 2027 (up from 56.7% and 19.3% in 2021 respectively); and
2. The core problems have their own dedicated targets for the broader missions.

Every year, MSIT reviews the performance of each Ministry's implementation plan. The review results are then reflected in the next year's implementation plan of the CP3. Performance management will be tailored to the characteristics of the social problem, depending if there are more appropriate technological, economic, or other metrics to measure success.

As part of the general Korean evaluation process for R&D programmes, the government conducts in-depth targeted evaluations (TEs). These evaluations cover, amongst other things, the alignment of R&D programs with national policy goals, their consistency with higher-level strategic plans, the appropriateness of budget allocation, and the effectiveness of the program's outcomes. The CP3 outlines a plan to conduct TEs for core social problems. If implemented, these evaluations would likely be performed by KISTEP under the commission of MSIT. The goal is to evaluate the appropriateness of mission-driven R&D programs (e.g., the adequacy of civic participation, relevant policies and systems, and government investment levels), as well as the linkage between R&D outcomes and the non-R&D elements necessary to achieve the goals (e.g., stakeholder involvement, institutional improvements, and cooperation with local governments). If a core social problem like 'cybercrime' is selected for TE, multiple R&D programs within that field become the subject of the evaluation, and programs are collectively assessed.

National Strategic Technology Policy

Another overarching mission-oriented innovation policy in Korea is the National Strategic Technology Policy (NSTP). It is a large-scale research and development programme that aims to support strategically

important technologies from the perspective of the economy, diplomacy, security, and new industry creation, with missions for each strategic technology. It takes a more systemic approach to supporting these technologies than traditional Korean STI policies have.

In 2021, the Korean government identified the Nation's 10 Critical and Emerging Technologies, which then were used as the basis for the NSTP under the next administration. Following the 2022 presidential elections, the new presidential programme agenda tasked MSIT to “establish a mission-oriented science and technology system to solve the problems facing the country such as carbon neutrality and aging, shift to private and local leadership, and strengthen industry-academic-research convergence and co-operation” (President of the Republic of Korea, n.d.^[12]). From this, the PACST Deliberative Council released a paper on how to design missions in October 2022, which was a major milestone for the NSTP. It included: announcing two new technologies from what the Strategy to Select, Promote, and Protect the Nation's Critical and Emerging Technologies had identified (for a total of 12); selecting 50 sub-technologies under the 12 National Strategic Technologies; targeting a 5-year R&D budget for the NSTP; establishing roadmaps (which were completed over 2023-24); and strengthening human resource development and international cooperation related to the NSTP.

The five-year S&T Basic Plan (2023-27) emphasises the role the NSTP will play to strengthen a mission-oriented R&D system, and strengthen the foundation for R&D creation, diffusion, utilisation and protection (MSIT, 2022^[13]). The commitment to this approach was further emphasised in the R&D Medium- to Long-term Investment Strategy for 2023-27, which places the NSTP and CNSTP at the forefront of four major strategies (MSIT, 2023^[14]). Legislation outlining the implementation of the NSTP was passed in March 2023 and enacted September 2023 – the Special Act on the Fostering of National Strategic Technology. According to the framework for mission designs, in the past, there has been insufficient national-level co-ordination on economic and societal missions with clear deadlines, for which science and technology can contribute to. This programme is the government's response to this gap (PACST, 2022^[17]).

In November 2023, the Korean R&D Innovation Plan and International R&D Promotion Strategy was released. It aims to foster world-leading innovative R&D in Korea and to leap forward to become a world-leading science and technology global hub. It is a joint strategy between MSIT, MOEF, MOTIE, and the Ministry of Education (MOE). The Plan emphasises the importance of the NSTP to drive long-term technology advancement, and it commits to providing the Policy with more budgetary flexibility for pan-ministerial work and international co-operation (MSIT, 2023^[26]). In August 2024, the PACST Deliberative Council released the NSTP Basic Plan, which announced an annual implementation plan will be delivered in 2025 for each department.

While there were broad cuts for other programmes, the Innovation Plan showed there would be continued support for the NSTP. The STI Office budgeted around 5 trillion won (~3.4 billion euros) in 2024 in strategic technologies, a 6.3% increase from 2023 when 4.7 trillion was budgeted (~3.2 billion euros). Through the Basic Plan, the target amount of R&D investment for the NSTP for 2024-28 has been increased from 25 trillion won to 30 trillion won (~17 billion euros to ~20 billion euros).

Strategic orientation

The NSTP provides direction to innovators and government ministries by identifying 12 strategic technologies, developing a roadmap for each one, and prioritising funding for programmes and projects that advance the roadmaps. For each of the 12 strategic technologies, the NSTP identified 2-8 sub technologies (for a total of 50), and then strategic roadmaps present time-bound and targeted missions for the technological capabilities to advance each sub-technology. Sub-committees for each of the 12 sub-technologies can recommend when their roadmaps need to be updated (KISTEP, 2023^[27]).

To determine the 12 strategic technology areas, the STI Office surveyed experts, consulted ministries, and had a peer review from industry, academia, and research institutes. PACST then reviewed and approved this work. Once the 12 strategic technology areas and 50 sub-technologies were decided upon, roadmaps with mission-oriented goals were developed for both the 12 strategic technologies and 50 sub-technologies (see Annex F for more information). This involved another round of consultations with the stakeholders, and relevant ministries.

Each roadmap includes a mission, and then outlines the R&D activities, and broader cross-government enabling policies (e.g., skills and regulations) needed to achieve the mission. They are focused on areas that can produce results over 5-7 years. For the 12 strategic technologies, the mission is often framed around being the top in the world for the technology. For each of the 12 strategic technologies, a roadmap provides interim milestones to achieve the final mission (see Box 4.3 for an example of how its missions are framed). For each of the 50 sub-technologies, there are more concrete technology-specific goals. One-page summaries are developed for stakeholders for each of the 50 sub-technologies to show the results of the roadmap, including the challenges, vision, missions, goals for technological development, and plans for ecosystem development.

Box 4.2. National Strategic Technology – example of mission framing

Challenge: Based on new digital services that overcome regional and spatial limitations, such as UAM, autonomous driving and digital twins, take the lead in the next-generation communications market through early commercialisation of 6G and the expansion of convergence services

Vision: Lead the 6G global market by securing standard patents and leadership in core technologies

Key technologies: 6G, 5G-Adv, satellite communications, Open-RAN, and high efficiency 5G-6G communication components

Example goals for technological development: Developing 6G original technology, securing standards and patents, strengthening the capabilities of the domestic mobile communication industry (Reflection of E-MIMO wireless communication technology standards); Strengthening industrial capabilities through the advancement of 5G wireless communication and network technology (base station power consumption reduced by more than 30% compared to 5G)

Policy strategy examples: talent/training, support for 6G international standardisation activities, and creating new institutions (e.g., 6G Forum and Open LAN Alliance)

Source: MSIT and KISTEP (2023^[28]), 국가전략기술 프로젝트 추진체계 구축 및 지원[Establishment and support of national strategic technology project implementation system], https://www.kistep.re.kr/reportDetail.es?mid=a10305030000&rpt_tp=831-002&rpt_no=RES0220230117.

The way this technology-focused mission is framed is closer to Korea's traditional technological priority exercises than those focusing on transformative societal goals. At the same time, by framing these technology priorities as missions, Korea is using new approaches for cross-government priority setting, and coordination bodies for developing programs that support these technologies more systemically. As a result, these technology-focused missions are helping reduce silos in the Korean STI system in a way that goes beyond traditional approaches.

Policy co-ordination

The missions aim to build an inclusive process throughout their lifecycle, and include the participation of experts from government, industry, academia, research institutes, and civil society in mission setting, R&D planning, and evaluation. This is formalised through a series of committees.

The National Strategic Technology Special Committee – a special committee under the PACST Deliberative Council – oversees the NSTP. The Special Committee, and 12 sub-committees focused on each strategic technology, are comprised of implicated government ministries, diverse experts from academia, research institutes, and industries. The main committee is chaired by the Vice Minister of Science and Technology Innovation Office.

The R&D and non-R&D elements of the roadmap are co-ordinated by the STI Office, which promotes co-operation between relevant ministries on cross-cutting R&D projects, as well as research institutes, academia and the private sector.

The STI Office also uses the budget process to coordinate the programmes from across government meant to support the NSTP. Through its reviews of new funding programmes, the STI Office prioritises proposals related to government strategies, of which the NSTP is a top priority. These considerations are based on the National Strategic Technology roadmaps, and looked at even more favourably if they include training and regulatory improvements in addition to investments in R&D.

The NSTP is also cited in several other sectoral strategies, notably the Basic Plan for Biotechnology Promotion (2023-32), and the 4th National Nanotechnology Roadmap (2023-32). Biotechnology is one of the 12 strategic technologies, and the sector-specific promotion plan discusses how it is complementary to the NSTP because the promotion plan will be more systemic. While nanotechnology is not a strategic technology or sub-technology area, several strategic technologies fund areas related to nanotechnology. The Nanotechnology Roadmap identifies how it complements strategic technologies under the NSTP and CNSTP. It also seeks to take a mission-oriented approach and sets a mission-oriented roadmap to address 'big questions' for the technology (PACST, 2023^[29]).

Box 4.3. Co-ordinating missions with structural funding

The US's Energy Earthshots Initiative was launched in 2021 by the United States. Department of Energy (DOE) with the aim of catalysing innovative advancements in various energy sector technologies to significantly reduce emissions in the United States and bring down the costs of select energy technologies. To achieve these reductions, eight Energy Earthshots have been established. They aim to achieve cost reduction and technical goals over approximately a ten-year period.

The Earthshots' policy mix includes research at DOE federal labs; funding of university, not-for-profit, and private research teams; funding the commercialisation of DOE research; challenge prizes; innovation for procurement; regulatory and permitting support; and technical assistance. Each Earthshot has a co-ordination team of both technical experts and leads from diverse teams that oversee investment tools within DOE. A science and innovation programme manager oversees each Earthshot co-ordination team, and the portfolio of projects and investments to tackle the mission.

The Earthshots co-ordinate with other initiatives depending on their priorities, and other strategies that affect their technology. For instance, the Hydrogen Shot has a joint strategy team under a broader Hydrogen Strategy with two co-chairs – one representing the DOE team responsible for the Earthshots and one from the DOE team responsible for infrastructure. This approach helps co-ordinate the various initiatives to support building out the infrastructure to deploy the technologies developed by the Earthshot. For example, the Bipartisan Infrastructure Law included \$9.5 billion for clean hydrogen initiatives and the Inflation Reduction Act creates a new Hydrogen Production Tax Credit. These initiatives all will advance the Hydrogen Shot and help bring the technologies it develops to market.

Another example is the Floating Offshore Wind Shot, which has taken an inter-agency approach, working alongside the Department of Transportation and Department of the Interior to set a roadmap and manage the Earthshot. This helps bring together the relevant regulatory reforms and accelerated permitting approvals that are needed to deploy the technologies developed by DOE's work.

Policy implementation

The NSTP mainly supports R&D to advance the mission areas, but the NSTP Basic Plan and roadmaps identify a range of other programmes including skills (including training but also easing visa requirements and immigration procedures), commercialisation, entrepreneurship support, clusters, industry-academia networking, innovation procurement, standard setting, and regulatory changes.

To implement the roadmaps, sectoral ministries are responsible for using their existing programmes, and developing new ones, to support the roadmaps. This includes R&D, but also the other types of programmes listed above. Government research institutes will also conduct R&D on core technologies. Funding for projects with private sector involvement will be largely based on the existing regulations, which has a R&D matching ratio of 50% for large companies, 30% for mid-sized companies, and 25% for small and medium-sized companies. The private investment ratio can be adjusted based on industry, field, scope of benefits, TRL, and spillover effects for the national economy. For instance, Strategic Research and Development projects, including flagship projects, can incentivize private investment through special benefits, such as reducing the R&D matching ratio for companies, lowering royalty fees, and providing support for follow-up projects.

From the 50 national strategic technologies, 10 flagship R&D projects were identified through a competitive process, and then provided with additional resources to enable their success. These flagship projects are strategically important and consist of a portfolio of projects that will be managed in a PM model, where the PM has discretion over the overall project (planning, evaluation, management, etc.) and manages the progress of the project. The government will provide additional resources to enable their success.

As is the case with all long-term Korean R&D projects, the projects will be evaluated to identify any opportunities and possible changes. The government will also establish a survey to ensure that an appropriate budget is available for promising projects. It will also be able to carry over unspent research funds to the next year for multi-year projects.

The Policy aims to employ a new process for evaluations suitable for missions, which focuses on both the approach and the projects. The government will review the strategy for each stage of the mission and monitor the performance as it progresses.

In the planning stage, it will seek to improve the pre-feasibility of mission achievement and strengthen internal capabilities through evaluations. These will also consider technical and societal changes surrounding the missions. When ministries establish strategic plans for mission-oriented R&D projects, specific performance indicators related to the mission are set and clear responsibility is assigned in a time-bound manner (e.g., quarterly achievements). These performance indicators can be used for planning, management, and evaluation of mission-oriented R&D projects.

When ministries establish strategic plans for mission-oriented R&D projects, specific performance indicators related to the mission are set and clear goals are assigned to ensure they are sufficiently targeting the mission area. Through assessments before and after the start of projects, it will also provide areas of improvement for the R&D projects and portfolios, taking into account social acceptability, systems, regulations, infrastructure, etc.

At the end of the Policy, (i.e., following mission achievement), an in-depth evaluation will be conducted on the performance, and technological and economic spillover effects of the missions. This analysis will be used to set the follow-up missions and improve the direction of government R&D project planning.

Box 4.4. The Carbon-Neutral Strategic Technology Policy

A parallel initiative to the National Strategic Technology Policy is the Carbon-Neutral Strategic Technology Policy (CNSTP).

It aims to reach Korea's 2030 nationally determined contributions to cut emissions and adapt to climate impacts under the Paris Agreement, and then achieve its 2050 carbon neutrality targets for each sector based on the national energy mix, global regulations, private sector reduction goals, etc.

The CNSTP has identified 17 carbon-neutral technologies, and 100 sub-technologies with accompanying mission-oriented roadmaps. Hydrogen is a technology that both the National Strategic Technology Policy and CNSTP are supporting.

The CNSTP is governed by:

- The PACST Deliberative Council, which is responsible for selecting core technologies, establishing technology roadmaps, and co-ordinating cross-ministerial R&D; and

- The Presidential Committee for Carbon Neutrality and Green Growth, which is responsible for improving consistency between the National Carbon Neutrality Policy and related policies, policy establishment, implementation, and supervision.

The carbon-neutral programming focuses on both technology development and deployment.

Source: KISTEP (2023^[30]), 탄소중립을 위한 혁신기술 [Innovative technology for carbon neutrality], https://www.kistep.re.kr/boardDownload.es?bid=0033&list_no=43710&seq=1.

The Alchemist

The Alchemist is a high-risk high-reward challenge-based programme that encourages the development of technologies in areas that have the dual purpose of improving Korean society and creating new sources of global leadership. It also seeks to tackle some of the weaknesses of the Korean R&D system, in particular, the relatively low appetite for pursuing risky projects (OECD, 2023^[11]). It is delivered by MOTIE's Korea Evaluation Institute of Industrial Technology (KEIT).

As discussed in Chapter 3, Korea's policy suite is undergoing a broad shift towards more challenge-led programmes. In May 2019, the MSIT convened a ministerial conference on science and technology development where relevant ministries agreed on the Measures to Strengthen National R&D Innovation, and the Alchemist Programme was nominated as a pilot project. It was one of several similar programmes that were piloted, including the Global-Future Innovation Research Science Technology project (G-FIRST), a collaborative programme by MSIT and MOTIE, and the Innovation Challenge Programme, another ARPA-style approach from MSIT (see Annex H for information on this program). However, both of these programmes failed their pre-feasibility tests.

The Alchemist's initial success has inspired other programmes that plan to use its methodology – notably the Defence Administration's Future Challenge Technology Development Project, the Ministry of Health and Welfare's ARPA-H and MSIT's Challenging the Limit Programme. Building on the success of the Alchemist Project, KEIT is currently planning follow-up challenge-oriented R&D initiatives, including projects like the Purple Ocean Project and the (tentatively named) Global Paradigm-Shifting Project.

Strategic orientation

The Alchemist programme sets themes, and then it supports a series of projects under each theme through a stage-gated approach – starting with six projects in year one of a theme, three projects in year two, and then one project is funded for five years. The Alchemist has set 19 themes so far (see Annex G). It will develop approximately three or four themes each year until 2025. Each theme is funded over a seven-year time-horizon. It has a budget of 414.2 billion won over 10 years (~280 million euros), starting in 2022, which will allow it to fund all of the stage gating for its final round of themes.

The programme uses a 'moving target approach', where each project under the theme is planned initially for seven years, but the projects are allowed to change as long as they still align with the themes. The programme also allows teams to merge if they find their research is converging, or for early termination if it is deemed that the project is not able to achieve its goal.

To set the themes, KEIT conducts desk research on the current state of technologies, and then what the future societal challenges will be. It also uses its big data platform to explore quantitative trends behind

technologies, and a public platform where citizens can submit their ideas. In addition, KEIT conducts a survey of top industry leaders and academics from the National Academy of Engineering, and other academics (PACST, 2021^[31]).

This work culminates in 20-30 draft themes. These draft themes are presented to the Alchemist Programme's Grand Challenge Committee, which consists of experts from science and technology, as well as humanities and social sciences, like science fiction writers, to help embed some futures thinking into the programme. Thematic working groups are created for each candidate theme to inform this work. The Grand Challenges Committee then selects 10-20 themes.

Once the challenge themes are selected, a working group of industry and academic experts then oversees each theme and decides what projects to fund.

Each theme has an accompanying brief that provides directionality by identifying several priority technology areas, and then how it envisions the innovations developed will improve society in the future. For instance, the Sustainable Beyond Plastics theme identifies bioplastics, new organic/inorganic materials, recycled plastic, and recycling technology as some related technologies. It then sets out its vision for the future as a "post-oil sustainable plastic economy and society, improving the functions of existing plastics. A society where material circulation of new plastics and composite materials based on innovative alternative renewable resources is possible" (Challenge Innovation Team, 2023^[32]). The briefs for each theme also include more technical analysis with regards to technical limitations, the innovativeness of the theme, prospects for creating new markets, expected timelines for the innovations to have societal impact, the global competitiveness in the technology areas, how Korea can secure global leadership in the field, the potential social impact, and technological differentiation.

Policy co-ordination

Three committees are involved in the governance of the programme, with different roles:

- The Programme Review Committee determined the pilot themes, and confirmed the scale and type of support. It consists of the chairman of the Grand Challenges Committee, government representatives, and third party experts;
- The Grand Challenge Committee – which includes representatives from science and technology, as well as humanities and social sciences – decides on new themes and performs consulting with stakeholders. It is informed by working of thematic working groups; and
- The KEIT Evaluation Committee evaluates results at each stage of the stage-gated funding, deliberates and adjusts the scale/budget/timeline for each theme.

Projects under each theme are managed by a team of about three dedicated Project Managers (PMs). This includes one PM from KEIT and two external experts.

KEIT has a dedicated team to support the planning, implementation and dissemination of the programmes. It also hosts inter-ministerial discussions to communicate with regulators and sector-specific departments on an ad-hoc basis, based on advances from the Alchemist projects. This has been particularly active in relation to health technologies.

Despite being within MOTIE, its planning is not deliberately aligned with MOTIE's or MSIT's technology and sector strategies (see Box 4.7. on how other countries have structured this).

Box 4.5. Overarching MOIPs using challenge-oriented policies in their policy mix

The United Kingdom's 2017 Industrial Strategy identified four overarching 'grand challenges' for the United Kingdom, with an accompanying set of targeted and timebound missions under the grand challenge areas. For instance, under the grand challenge related to ageing society, it set a mission to "ensure that people can enjoy at least five extra healthy, independent years of life by 2035, while narrowing the gap between the experience of the richest and poorest". This now withdrawn strategy was an overarching mission-oriented innovation policy.

The grand challenges were supported by a suite of initiatives. The healthy ageing mission established a Longevity Council to help the United Kingdom seize the economic opportunities of ageing societies, funded healthcare innovations through the UKRI's Strategic Priorities Fund, and launched a joint United Kingdom–Japan competition to support British and Japanese businesses to harness AI and robotic assisted living products.

Each mission was also supported by the Industrial Strategy Challenge Fund (now the UKRI Challenge Fund), which is a more challenge-oriented policy and its work has continued even after the grand challenges were withdrawn. For the Challenge Fund, a challenge director oversees a suite of collaborative research projects between businesses and other firms or academia. It set 23 challenges, three of which were created for the ageing society challenge:

- Accelerating detection of disease - Building a national resource of volunteers to support the development of new diagnostic tools and technologies
- Healthy ageing - Developing products and services that help people remain independent and active into older age
- Data to early diagnosis and precision medicine - Supporting the development of precision medicine for improved early diagnosis and treatment

This nested structure, where an overarching mission—oriented innovation policy uses a challenge-oriented policy to help in the delivery, is also being used in the United States' Energy Earthshots Initiative. Here the United States Advanced Research Projects Agency–Energy (ARPA-E) (a challenge-based organisation) advances high-risk, high-reward research under the mission.

Policy implementation

Non-profits or research organisations are leaders for each of the themes to oversee the work from a technical perspective. The participants in the programme are found by the research agencies. Themes are managed by a dedicated group of about three Project Managers who oversee the projects step-by-step and review their progress. Individual organisations and consortia then apply for funding, which are often led by non-profits or research organisations. A company must be involved in every project team to encourage commercialisation, and technology transfer of the intellectual property they create. There is a suggested government cost sharing rate for the participants: large companies (<50%), medium-sized companies (70%), small companies (<75%), and research institutions (<100%). The cash ratio of contributions should be not less than 30% for all types of companies. There are additional incentives for

companies by lowering the matching ratio depending on size: increasing of government support rate (75% to 80%, for small sized companies only); and reducing the cash ratio of private contributions (from 30% to 10%, for Small and Medium sized companies only). There is no requirement for how much universities and public research institutions should invest.

KEIT selects the winning projects for each round of the contest, based on the advice of the Grand Challenge Committee and the designated research institute involved in overseeing the theme. This repeats through the three-stage competitive process:

- Proof of concept research – 1 year, 6 projects, 200 million per project (~140,000 euros)
- Pilot research – 1 year, 3 projects, 500 million won per project (~340,000 euros)
- Main research – 5 years, 1 project, 4 billion won per year (~2.7 million euros)

KEIT tracks and manages project teams that do not pass onto the next round, and these teams can access other forms of support from MOTIE instead – including for commercialisation or further R&D. KEIT operates its own programmes to encourage follow-on R&D and commercialisation from technologies developed under the programme.

The Alchemist's slogan is “we accept failure”, as it wants researchers to take risks and not be afraid of failure if they do not succeed or do not make it to the final round of funding. This approach is in contrast to other R&D programmes delivered by MOTIE, where there are penalties for researchers who do not achieve their goals – such as having to repay part of the funding they received – which discourages researchers from pursuing ambitious and high-risk projects.

For all the project teams, KEIT provides IP support, including patent lawyers to carry out analysis on global patent trends, and to coach each project on how to increase the value of their IP. It also offers networking between the members to encourage the dissemination of ideas, and once a year the participating researchers and members under each theme hold a forum to talk about the project.

The Alchemist has a special membership programme, where companies who do not participate in the R&D projects directly can monitor their progress for free through the first two stages of the programme, and then pay a subscription fee at the third stage to gain priority access to the IP. These members sometimes also provide facilities to the research teams. The membership system encourages competitors to be involved in like-minded projects, and both benefit at the end.

For the programme at large, the stage-gated approach builds in opportunities to monitor progress towards the goal. The Evaluation Committee evaluates results at each stage of the stage-gated funding, deliberates and then adjusts the scale, budget, and timeline for each theme. Instead of determining if a project is a success or failure based achieving the goal alone, a new evaluation system was established to factor in industrial impact (PACST, 2021^[31]).

The programme has been such a successful model for supporting innovation to date that other ministries and regional governments are creating their own versions. KEIT is helping them by sharing their corporate documents.

5

Conclusions and options for the future of the mission-oriented policy agenda in Korea

Korea's STI institutional setting is conducive to mission-oriented policies

Korea has a long-established strategic agenda setting model for giving its STI system direction, which includes 10-year, five-year, and annual STI action plans, as well as sector- and technological-specific ones. This approach already provides Korean innovators with a level of directionality that is above and beyond what many countries offer. It is clear by the sheer number of mission-oriented programmes, and more technology-driven challenge-oriented ones, that the country is committed to providing even more directionality to its innovation system. In particular, a number of mission-oriented plans have been put in place, particularly since 2018, to prioritise technological development that benefits Korean society and drive competitiveness. While Korea has several different MOIPs, this study focuses on three: the 3rd Comprehensive Plan to Solve Social Problems Based on Science and Technology (2023-27) (CP3), the National Strategic Technology Policy (NSTP), and the Alchemist (highlighted in Chapter 4). This provides a representative sample of the different types of MOIPs Korea is using. Annex I summarises the main characteristics of these MOIPs.

Leading foresight capabilities are another strength in the Korean STI system, which helps the government to identify what issues matter most to Koreans, and what technology and societal trends are emerging, which may in turn require a government response. The use of big data analytics for identifying thematic areas under the Alchemist programme, and the use of citizen sentiment analysis for identifying priority social problems under the CP3, are two examples of how this is already being applied in a MOIP context. By taking such an evidence-based approach to these priority-setting exercises, Korea's priorities are sometimes able to transcend political cycles, where other countries again have struggled to maintain momentum from one election to another. Again, the CP3 is a testament to this, where only four of the 43 social problems are new under the 3rd plan compared to the 2nd (see Annex C).

In addition, the STI Office's strong co-ordinating role, including in the budgeting process, ensures there is a moderate level of coherence across the government's STI policies, and R&D projects. While it has traditionally focused on how to prevent duplication of STI supports across government, MOIPs are providing new opportunities for the government to collaborate on identifying and addressing gaps in the

STI support system to help achieve their missions. This co-ordination has been formalised through new committees overseeing each MOIP initiative, and then at each mission.

The MOIPs have mapped these programmes to strategic roadmaps on how they plan to reach each mission area. This includes a diverse set of policy levers, including R&D investments in key areas, skills development, regulatory sandboxes, and procurement for innovation. New programmes that address missions under the NSTP and CNSTP have been prioritised through Korea's budgeting process, which has mobilised government-wide resources. In addition to funding for R&D, these mission-oriented innovation policies are deploying a range of policy instruments that even initiatives within Korea could learn from each other. This includes changing the rules behind the funding (i.e., removing technology fees and not penalising failure), and using more demand-side instruments (e.g., using existing procurement of innovation programmes or granting regulatory sandboxes for technologies developed through some of the MOIPs). The Alchemist's challenge-oriented approach has also been recognised as a successful approach for driving technological advancement to address societal challenges. In contrast to the overarching-MOIPs, it is more focused purely on technology development. It uses a stage-gated approach to funding and better embraces failure than traditional STI programmes. Finally, recent changes to monitoring and evaluation of STI policies is also allowing the government to consider the collective impact from programmes, as work towards missions is the sum of many parts.

Options for change

Korea has engaged in reforms and strategic initiatives that have embedded MOIPs in its national system of innovation. Notably, it has used the mission-oriented NSTP and the CNSTP as central pillars for the five-year S&T Basic Plan, and included them in its nested set of strategies to deliver S&T programming. In addition, the CP3 has a mission layer to concentrate work on addressing five core societal issues, which goes beyond what previous iterations took to provide directionality. The CP3 is also included in the five-year S&T Basic Plan. This work is in addition to a series of ARPA-type challenge programs the government has launched to drive ambitious technological change (e.g., the Alchemist launching challenges on areas such as reverse aging).

These policies have added more directionality to the Korean innovation system and are helping to encourage more transformative technological progress and societal change. At the same time, they are all relatively new and will require time to establish themselves before being evaluated in depth.

To help embrace the transformative nature of missions beyond the strong foundation in place, the OECD proposes one overarching recommendation for setting future missions in the mid- to long-term (Recommendation 1) and several additional recommendations to improve existing missions in Korea in the coming years (Recommendations 2 to 8). The achievement and lessons learned in the second set of shorter-term recommendations will set the conditions for the implementation of the Recommendation 1.

Overall approach for future missions

Recommendation 1. Set a small number of transformative missions for societal challenges with clear, inspiring and powerful directions at the highest level of government to galvanise national support. Korea's significant steps towards a mission-oriented approach to date have provided more directionality and are encouraging more collaborative approaches across government. These missions are, for the most part, technological in nature. While new technologies are essential for solving societal challenges such as global warming (IEA, 2023^[33]), wider changes that combine technological, behavioural and societal changes, will be necessary to successfully conduct the necessary transitions and create

nation-wide buy-in. To that effect, Korea could build on its experience of mission-oriented innovation policies to launch a smaller number of missions with clear, inspiring and powerful directions, rooted in the highest level of government. Other more technology focused initiatives such as the NSTP and the CNSTP that are central pillars of the five-year S&T Basic Plan, along with ARPA-type challenge programs the government has launched to drive ambitious technological change, would contribute to these larger and higher-level missions.

Anchoring whole-of-government, transformative, missions above the traditional STI strategic and policy frameworks such as Basic Plan is a challenge in many countries. Options include creating an executive body at the centre of government (e.g. missions led by the SGPI under the Prime minister office in France) or mandating a high-level cross-ministerial body (e.g. missions managed by a high-level Committee in Japan, with strong support from the Cabinet Office). The United-Kingdom has established five overarching national missions to drive the interventions of its “mission-driven government” elected in July 2024. A cross-ministerial platform taking responsibility of the missions, based on their broader remit and stronger systemic responsibilities across the government structure, a dedicated agency linked to several ministries or a large public-private partnership/platform or an autonomous organisation with foundation status could also lead such missions (Larrue, Tönurist and Jonason, 2024^[34]).

Strategic orientation of current missions

Recommendation 2. Raise the profile of Korea’s societal missions to encourage more support for them and allow for a more systemic approach at the national level. Korea has a well-established set of processes and strategies for identifying government STI priorities and encouraging sectoral ministries to align their supports – notably the 5-year STI Basic Plan, the R&D Medium- to Long-term Investment Strategy, and the annual programme review and budgeting process. The NSTP and CNSTP are prioritised in these processes and strategies. Furthermore, their integrated governance and mission-oriented roadmaps builds even more coordination towards their priorities than traditional policies. This approach provides these policies with a high degree of status and ensures there is buy-in to deliver on their accompanying technology priorities from across government. The CP3 does not have the same profile, and sometimes faces challenges to build the same level of buy-in as these other MOIPs. While countries have different perspectives on the importance of policies elevating the CP3 to be at the core of the five-year S&T Basic Plan and prioritised through its budgeting processes (e.g., the five-year Investment Strategy), alongside the NSTP and CNSTP, would further encourage buy-in to achieve transformative societal change.

Elevating the status of the CP3 missions will be particularly instrumental in the lead up to the definition of the whole-of-government missions proposed in the Recommendation 1.

Recommendation 3. Create more co-ordinated priority setting amongst its various ARPA-style challenge programmes to complement the overarching mission-oriented frameworks. Some other countries have used a nested approach where challenge-oriented programmes have supported broader overarching mission-oriented innovation programmes (e.g., ARPA-E being one tool supporting the Energy Earthshots in the US). The various ARPA-style programmes Korea has underway and under development could look to align some of their technology priorities with that of the NSTP, CNSTP, and CP3. While the aim is not to align them perfectly, this could help leverage areas for complementarity and would bring more consistency overall.

Implementation of current missions

Recommendation 4. Establish a dedicated portfolio management structure and related practices to provide proactive and holistic steering of set of activities contributing to the missions. Missions are similar to programmes, as they have boundaries regarding what projects and initiatives are helping achieve their goals. ARPA-style challenge programmes (e.g., the Alchemist) have narrower boundaries with a programme managers assigned to each challenge area to manage the portfolio of projects. This role involves monitoring progress towards collectively-developed roadmaps, directing resources towards promising lines of research and away from those with less promise, and identifying complementarities between projects so they can collaborate to achieve the mission goals. Portfolio management for overarching mission-frameworks requires similar levels of oversight but at a much larger scale. The governance and implementation following the programme approval stage in Korea could go further in ensuring projects and initiatives are more aligned towards the goals through a dedicated portfolio management approach. This hands-on support aligning, monitoring and steering activities to advance the mission can build momentum and help bring more coherence to individual programmes and projects that are often monitored individually across different ministries.

Recommendation 5. Provide societal missions with flexible, dedicated, multiyear funding to embrace their transformative potential. In providing a dedicated, multi-year budget for missions, governments demonstrate their priorities. This provides stakeholders with confidence to invest in these areas. MOIPs also require strategic and operational flexibility in how they fund projects and initiatives within their portfolio, as the path to achieve their mission goals may evolve and require different types of interventions over time. Some countries (e.g., France 2030 – see box 3.2) use a central pot of funding that can be distributed to various activities (R&D and non-R&D activities) in support of the missions across different missions and ministries (horizontally), as well as other funding for the work of specific missions and ministries (e.g., vertically). A case for a mix of vertical and horizontal funding was made in Korea's 5th Science and Technology Basic Plan 2023-27. The CP3 also seeks to establish a new budgeting system as one of its ten key tasks that can be used to fund initiatives across the social problems, instead of relying as heavily on contributions from the sectoral ministries. In addition, while the National Fiscal Act does not allow Korea to implement multi-year budgeting at this time, its 5-year R&D Medium- to Long-term Investment Strategies provides notional benchmarks for how much funding will be dedicated towards government priorities. The latest Investment Strategy sets a benchmark to fund the NSTP and CNSTP and a similar commitment could be made for the societal missions under the CP3.

Recommendation 6. Align STI support along the innovation chain from across government to improve missions' chances of success. There are many different technology priority exercises across the Korean government, which are used to provide preferential government funding and other supports at different points along the innovation chain (e.g., R&D tax credits from the MOEF, and preferential direct funding for applied R&D and commercialisation from MOTIE). While there is a large overlap amongst these priorities, they are not perfectly aligned and many MOIPs (e.g., MSIT's NSTP) only have some of their missions qualify for preferred support. Aligning these technology priority exercises and the supports that target different stages of technology development would ensure more types of programmes are available for each MOIP along the innovation chain. This alignment between the mission-oriented programs and other departments will be particularly important as missions mature and focus more on technology deployment than development.

Recommendation 7. Develop pathways to market in current and future challenge-oriented programmes to realise the full impact of the technologies they develop. Embedding more procurement and regulatory mechanisms into the challenge-oriented policies would better steer innovation and create new markets for the technologies developed under the challenges. A major reason for the

success of other international ARPA programmes is the use of government procurement to help bring these products to market. The CP3 and NSTP have embedded preferential public procurement for innovations developed under their programs. These practices could serve as best practices for the Alchemist and other ARPA initiatives currently under development.

Recommendation 8. Adopt mission monitoring and evaluation processes and practices that capture their specific systemic effects and assess their added value vis-a-vis traditional STI policies. The evaluation of MOIPs must explore the extent to which the additional work to develop and co-ordinate in a mission-oriented manner brings any benefit beyond what traditional strategies offer. Korea can build on recent improvements to its monitoring evaluation system to capture these more systemic elements and strategic aspects of MOIPs in its evaluations. Recent OECD work also provide evaluation frameworks and tools tailored to assess missions' specific additionality (Larrue, Tönurist and Jonason, 2024^[34]).

References

- Ahn, S., K. Lee and S. Lee (2021), "Major Issues and Policy Recommendations for Government R&D Subsidies for SMEs", in *KISTEP Issue Paper 2021-04*. [42]
- Appelt, S. and F. Galindo-Rueda (2023), "Working Party of National Experts on Science and Technology Indicators: OECD R&D tax incentives database, 2022 edition", OECD Publishing, Paris. [40]
- Challenge Innovation Team (2023), "지속가능한 비욘드 플라스틱 [Sustainable Beyond Plastic]". [32]
- Eum, S. (2023), "Conversion of the green mission for transformation-oriented industrial innovation: Green Growth and 2050 Carbon Neutrality in South Korea", *SocArXiv*, <https://doi.org/10.31235/osf.io/2narz>. [7]
- European Commission (2018), "Mission-Oriented Research and Innovation Inventory: and characterisation of initiatives", <https://doi.org/10.2777/697082>. [2]
- European Commission (2018), "Mission-Oriented Research and Innovation: Assessing the impact of a mission-oriented research and innovation approach", <https://doi.org/10.2777/373448>. [3]
- Fine Dust Special Countermeasure Committee (2019), "미세먼지 관리 종합계획 (2020-2024) [Fine dust management comprehensive plan (2020-2024)]", https://www.airkorea.or.kr/portal/web/board/9/1565/?pMENU_NO=159&page=7. [48]
- Green Growth Committee (2009), "녹색성장국가전략 [Green Growth National Strategy]", http://www.climate.go.kr/home/cc_data/policy/greengrowth_strategy.pdf. [8]
- IEA (2023), "Net Zero Roadmap: A Global Pathway to Keep the 1.5 °C Goal in Reach", IEA, Paris, <https://www.iea.org/reports/net-zero-roadmap-a-global-pathway-to-keep-the-15-0c-goal-in-reach>. [33]

- KISTEP (2023), “2023 년도 혁신도전 프로젝트 사업 연구기획과제 공모(제 2 차) [2023 Innovation Challenge Project Business Research Planning Project Contest (2nd)]”, <https://www.kistep.re.kr/flexer/view.jsp?FileDir=/board/0028&SystemFileName=202307130927563731.hwp&ftype=hwp&FileName=202307130927563731.hwp>. [51]
- KISTEP (2023), “국가전략기술 프로젝트 추진체계 구축 및 지원 [Establishment and support of national strategic technology project implementation system]”, https://www.kistep.re.kr/reportDetail.es?mid=a10305030000&rpt_tp=831-002&rpt_no=RES0220230117. [27]
- KISTEP (2023), “탄소중립을 위한 혁신기술 [Innovative technology for carbon neutrality]”, https://www.kistep.re.kr/boardDownload.es?bid=0033&list_no=43710&seq=1. [30]
- KISTEP (2021), “2020 년도 사회문제해결 R&D 투자 현황 및 특징 [Investment trend in social problem-solving R&D in 2020]”, https://www.kistep.re.kr/boardDownload.es?bid=0031&list_no=42217&seq=1. [37]
- KISTEP (2021), “2021 년 과학기술혁신정책지원사업 도전적·혁신적 R&D 촉진을 위한 법제 개선 연구 [2021 Science and Technology Innovation Policy Support - Project Research on improving legislation to promote challenging and innovative R&D]”, https://www.kistep.re.kr/reportDetail.es?mid=a10305030000&rpt_tp=831-002&rpt_no=RES0220220037. [22]
- KISTEP (2021), “2021 년 혁신도전 프로젝트 사업 지원 [2021 Innovation Challenge Project Support]”, <https://www.k2base.re.kr/ekp/view.do?templnt=341&vStartP=393&rptNo=RES0220220120&rptTp=%EC%A0%95%EC%B1%85%EA%B8%B0%ED%9A%8D>. [49]
- KISTEP (2021), “A study on the basis for facilitating multi-ministerial joint planning in 2021”, https://www.kistep.re.kr/reportDetail.es?mid=a10305090000&rpt_no=RES0220220106. [46]
- KISTEP (2020), “2019 년 과학기술종합조정지원사업 - 미션 중심 도전적 R&D 프로그램 기획 연구 [2019 Science and Technology Comprehensive Co-ordination Support Project - Mission-oriented challenging R&D programme planning research]”. [10]
- KISTEP (2020), “2020 년 혁신도전 프로젝트 사업 지원 [2020 Innovation Challenge Project Support]”, <https://www.k2base.re.kr/ekp/view.do?templnt=341&vStartP=393&rptNo=RES0220210149&rptTp=%EC%A0%95%EC%B1%85%EA%B8%B0%ED%9A%8D>. [23]
- KISTEP (2019), “2019 1st National R&D Programme Specific Evaluation Report - Multiministerial programmes”. [45]
- KISTEP (2019), “Study on Direction for Improving Governmental R&D Investment through Company Support”, <https://scienceon.kisti.re.kr/commons/util/originalView.do?cn=TRKO202000024287&dbt=TRKO&m=>. [43]

- Koo, B. (2023), “기업 혁신활동 제고를 위한 R&D 조세 지원 정책 연구 : 국가전략기술 연구개발 기업을 중심으로 [Research on R&D tax support policy to enhance corporate innovation activities: Focusing on national strategic technology research and development companies]”, https://www.kistep.re.kr/board.es?mid=a10306010000&bid=0031&list_no=43311&act=view. [41]
- Larrue, P. (2021), “The design and implementation of mission-oriented innovation policies: A new systemic policy approach to address societal challenges”, in *OECD Science, Technology and Industry Policy Papers*, OECD Publishing, Paris. [5]
- Larrue, P. (2021), “The design and implementation of mission-oriented innovation policies: A new systemic policy approach to address societal challenges”, *OECD Science, Technology and Industry Policy Papers*, No. 100, OECD Publishing, Paris, <https://doi.org/10.1787/3f6c76a4-en>. [6]
- Larrue, P., P. Tönurist and D. Jonason (2024), “Monitoring and evaluation of mission-oriented innovation policies: From theory to practice”, in *OECD Science, Technology and Industry Working Papers*, OECD Publishing, Paris, <https://doi.org/10.1787/5e4c3204-en>. [34]
- Lee, H. et al. (2023), “임무중심 국가과학기술혁신정책 전환방안 연구 II, A Study on Transitioning to a Mission-centered National STI policy II”. [50]
- Mazzucato, M. (2018), “Mission-Oriented Research and Innovation in the European Union - A problem-solving approach to fuel innovation-led growth”, Directorate-General for Research and Innovation, European Commission, <https://doi.org/10.2777/36546>. [4]
- Ministry of Environment (2020), “지속가능한 녹색사회 실현을 위한 대한민국 2050 탄소중립 전략 [Korea's 2050 Carbon Neutral Strategy to Realize a Sustainable Green Society]”, https://www.kcen.kr/USR_main2016.jsp??=data/data04_view&bbs_no=15011&page=1&bbs_option_cd=&search_option=0&search_value=&h_search=. [9]
- Ministry of SMEs and Start-ups (2021), “Directions for SMEs Success”. [44]
- MOTIE (2022), *국가첨단전략산업 경쟁력 강화 및 보호에 관한 특별조치법 [Act on Special Measures for Strengthening and Protecting the Competitiveness of National High-tech Strategic Industries]*, <https://www.law.go.kr/LSW/lsInfoP.do?lsId=014238&ancYnChk=0#0000>. [16]
- MOTIE (2021), “국가핵심기술 69 개에서 71 개로 확대 지정 국가핵심기술 보호조치사항에 대한 세부지침 마련 [Designation of national core technologies expanded from 69 to 71. Detailed guidelines for national core technologies protection measures established.]”, <https://www.korea.kr/briefing/pressReleaseView.do?newsId=156431942>. [15]
- MSIT (2024), *2024 년 정부연구개발 예산의 주요 특징 [Key features of the 2024 government research and development budget]*. [18]
- MSIT (2023), ‘윤석열 정부 R&D 혁신방안’, ‘글로벌 R&D 추진전략’ 발표 [‘Yoon Suk-yeol Government’s R&D Innovation Plan’ and ‘Global R&D Promotion Strategy’ Announced], <https://www.korea.kr/briefing/policyBriefingView.do?newsId=156601825>. [26]

- MSIT (2023), 제 1 차 국가연구개발 중장기투자전략(2023-2027) [*R&D Medium- to Long-term Investment Strategy for 2023-27*], https://www.kistep.re.kr/reportDetail.es?mid=a10305040000&rpt_tp=831-005&rpt_no=RES0220230098. [14]
- MSIT (2022), 제 5 차 과학기술기본계획 2023-2027 [*The 5th Science and Technology Basic Plan, 2023-2027*], https://www.kistep.re.kr/reportDetail.es?mid=a10305030000&rpt_tp=831-002&rpt_no=RES0220230107. [13]
- MSIT (2020), 대한민국 과학기술 미래전략 2045 미래를 향한 도전과제와 과학기술정책의 전환 [*Korea Science and Technology Future Strategy 2045 Challenges for the future and transformation of science and technology policy*], <https://scienceon.kisti.re.kr/srch/selectPORSrchReport.do?cn=TRKO202000029214>. [11]
- MSIT and KISTEP (2024), 2022 년도 연구개발활동조사 [2022 R&D Activity Survey], https://www.kistep.re.kr/reportDetail.es?mid=a10305060000&rpt_tp=831-001&rpt_no=RES0220240054. [39]
- MSIT and KISTEP (2023), 국가전략기술 프로젝트 추진체계 구축 및 지원[*Establishment and support of national strategic technology project implementation system*], https://www.kistep.re.kr/reportDetail.es?mid=a10305030000&rpt_tp=831-002&rpt_no=RES0220230117. [28]
- MSIT and MOEF (2024), R&D 예타 조사 전면 폐지...‘사전 전문검토’ 등 도입 [Abolition of R&D Preliminary Surveys Completely...Introduction of ‘Preliminary Professional Review’], <https://www.korea.kr/news/policyNewsView.do?newsId=148929908>. [38]
- OECD (2024), *Main Science and Technology Indicators (database)*, <https://www.oecd.org/en/data/datasets/main-science-and-technology-indicators.html>. [20]
- OECD (2024), *OECD Science, Technology and R&D Statistics (database)*, https://www.oecd-ilibrary.org/science-and-technology/data/oecd-science-technology-and-r-d-statistics_strd-data-en. [36]
- OECD (2024), *OECD Tax Incentives Database (database)*, [https://data-explorer.oecd.org/vis?df\[ds\]=dsDisseminateFinalDMZ&df\[id\]=DSD_RDTAX%40DF_RDTAX&df\[ag\]=OECD.STI.STP&df\[vs\]=1.0&pd=2015%2C&dq=.A..PT_B1GQ..&to\[TIME_PERIOD\]=false&ly\[rw\]=REF_AREA&ly\[cl\]=TIME_PERIOD&ly\[rs\]=MEASURE](https://data-explorer.oecd.org/vis?df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_RDTAX%40DF_RDTAX&df[ag]=OECD.STI.STP&df[vs]=1.0&pd=2015%2C&dq=.A..PT_B1GQ..&to[TIME_PERIOD]=false&ly[rw]=REF_AREA&ly[cl]=TIME_PERIOD&ly[rs]=MEASURE). [21]
- OECD (2023), *OECD Reviews of Innovation Policy: Korea 2023*, OECD Reviews of Innovation Policy, OECD Publishing, Paris, <https://doi.org/10.1787/bdcf9685-en>. [35]
- OECD (2023), *OECD Reviews of Innovation Policy: Korea 2023*, OECD Reviews of Innovation Policy, OECD Publishing, Paris, <https://doi.org/10.1787/bdcf9685-en>. [1]
- OECD (2023), *OECD Science, Technology and Innovation Outlook 2023: Enabling Transitions in Times of Disruption*, OECD Publishing, OECD, <https://doi.org/10.1787/0b55736e-en>. [52]
- PACST (2023), 제 3 차 과학기술 기반 사회문제해결 종합계획 (2023-27) (안) [3rd Science and Technology-Based Comprehensive Solution Plan for Social Issues (2023-27) (Draft)], https://www.pacst.go.kr/jsp/council/councilArchiveView.jsp?archive_id=1070&cpage=3. [25]

- PACST (2023), 제 4 기 국가나노기술지도(2023-32)(안) - 나노기술 혁신을 통한 새로운 전환 [*The 4th National Nanotechnology Map (2023-32) (draft) - New transition through nanotechnology innovation*], https://www.pacst.go.kr/jsp/council/councilArchiveView.jsp?archive_id=1107&. [29]
- PACST (2022), 국가적 난제 해결을 위한 임무중심 R&D 혁신체계 구축전략(안) [*Strategy for establishing a mission-oriented R&D innovation system to resolve national challenges (draft)*], https://www.pacst.go.kr/jsp/council/councilArchiveView.jsp?archive_id=28&cpage=8. [17]
- PACST (2021), 제 7 차 산업기술혁신계획(2019-23) 2021 년도 시행계획(안) [*7th Industrial Technology Innovation Plan (19-23) 2021 Implementation Plan (draft)*], https://pacst.go.kr/jsp/m/council/councilPostView.jsp?post_id=2095&board_id=11&etc_cd1=COUN01#this. [31]
- President of the Republic of Korea (n.d.), 120 대 국정과제 [*120 National Policy Tasks*], https://www.president.go.kr/affairs/gov_project. [12]
- Related Joint Ministers (2024), R&D 선순환 촉진을 위한 기술료 제도개선 방안 [*Measures to improve the technology fee system to promote a virtuous cycle of R&D*]. [19]
- Related Joint Ministers (2016), “국민 안전과 건강 보호를 위한: 미세먼지 관리 특별대책 [Special measures to manage fine dust to protect public safety and health]”. [47]
- STI Office (2006), 비전 2030 실현을 위한 기술기반 삶의 질 제고방안 (요약) [*Measures to improve quality of life based on technology to realize Vision 2030 (Summary)*], <https://archives.knowhow.or.kr/policy/presidentialReport/view/20593>. [24]

Annex A. Detailed framework conditions for mission-orientation in Korea

This annex provides a synthesis of the evolution of the broader Korean STI governance system, which is discussed in more detail in the OECD Review of Korean innovation policy (OECD, 2023^[1]).

The development of the Korean STI governance system

There are three main periods for how Korea's STI governance has evolved:

1962-2000: The emergence of an autonomous national innovation system – Korean STI policy in this period was guided closely by five-year national economic development plans that focused on acquiring and learning from foreign technologies in targeted industries. Government research institutes (GRIs) played a major role in absorbing key foreign technologies and transmitting them to large Korean conglomerates. Private firms also received significant guidance and support from state authorities via subsidised loans, specific licences to enter targeted industries and protective regulations.

In the 1980s, more autonomous STI policies emerged to incubate national S&T capacity, encourage the formation of industry consortia, and promote the development of core technologies, such as semiconductors and various information technology (IT) technologies. This approach was then applied in other sectoral ministries.

In the 1990s, the focus turned to fundamental science and its commercialisation. The first mid- to long-term national R&D plan was released in 1992 and is seen in retrospect as being instrumental in fostering industries where Korean companies are now global leaders: semiconductors, various IT segments, electronics and automobiles. This period also saw an increase in complexity for the Korean STI support system, as the scale and scope of STI support expanded rapidly across sectoral ministries (agriculture, transport, health, land and construction, etc.) that started to invest significantly in R&D activities at this time.

2001-22: The maturing of the STI governance system – This period started with a radical change in the structure of STI governance to respond to the challenges of co-ordination across the Korean government and the more robust suite of supports for STI, which had to be planned, managed and evaluated. New frameworks and organisational structures were established at this time, including the creation of the Ministry of Science and Technology in 1998; the establishment of a new, more powerful, co-ordination council in 1999 (the National Science and Technology Council, NSTC); the enactment of the Framework Act on Science and Technology in 2001; the launch of the first five-year S&T Basic Plan in 2002; and the creation of an executive office to take charge of an increasing range of STI governance functions (notably, the Science, Technology and Innovation Office; hereafter).

There were also attempts to merge or reorganise ministries and agencies to improve co-ordination and/or streamline structures. For instance, in 2008, the Ministry of Science and Technology (MOST) and the Ministry of Education (MOE) merged to become the Ministry of Education, Science and Technology (MEST). Also significant was the creation in 2008 of the Ministry of Knowledge Economy (MKE), integrating parts of the former Ministry of Commerce, Industry and Energy (MOCIE), the Ministry of Information and Communication (MIC) and MOST. The aim of this “super ministry”, active until 2013 when it was disbanded,

was to have a more integrated policy structure and to increase co-operation between policy portfolios to strengthen the knowledge and technological base of industries in key industries to develop new sources of growth. The government granted a powerful mandate to central executive and co-ordination bodies, and introduced an increasing number of co-ordination, budgeting and monitoring mechanisms, rules and guidelines throughout the entire government structure.

2022 onwards: Towards a new STI governance system – Korea is currently entering a third stage of STI governance, which sees it using its STI system to prioritise specific societal challenges – including health/aging, and climate change – through co-ordinated, whole-of-government strategies. While this approach can trace its origins back to 2006/07 with the Technology-based Quality of Life Enhancement Plan for Vision 2030, and Comprehensive Measures to Improve the Quality of Life Based on Technology, these documents did not result in fundamental changes to how the Korean government operated but represented some early attempts to direct research towards societal priority areas. A new series of reforms starting in 2022 is helping address these challenges by taking a whole-of-government mission-oriented approach (see chapter 4 for more information).

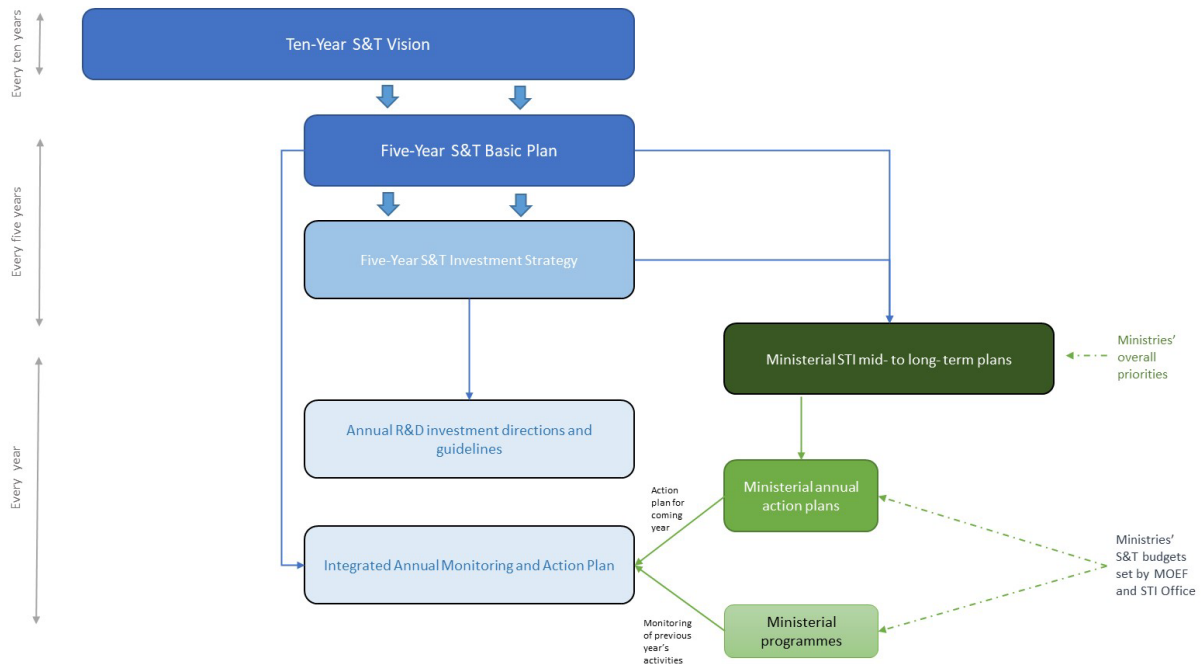
Strategic Orientation

This section explores the first dimension of mission-oriented policies, strategic orientation, in the Korean context. It includes the relevant strategies, priority-setting activities and trends in the Korean STI system and how effective they are in setting directions that concretely influence the various actors' own agendas and actions.

Overarching STI agenda-setting in Korea

The Korean government has a cascading system of well-elaborated whole-of-government strategies for providing long-term direction. For encouraging strategic orientation in Korean STI policy, it has several long-standing mechanisms (as outlined in Figure A.1): the long-term vision, the presidential programme agenda, the five-year S&T Basic Plan, and annual plans for each ministry. Ad-hoc strategies complement this work.

Figure A.1. Korea's STI governance system: From long-term strategic orientations to annual programming and budgeting



Source: OECD (2023^[35]), *OECD Reviews of Innovation Policy: Korea 2023*, OECD Reviews of Innovation Policy, OECD Publishing, Paris, <https://doi.org/10.1787/bdcf9685-en>.

The long-term STI visions: Korea establishes long-term visions for S&T about every ten years. Previous visions include the Long-term Vision for S&T Development by 2025, launched in 1999, and the S&T Future Vision for 2040, launched in 2010. The current long-term vision from 2020, *Innovate KOREA 2045 – Challenges and Changes for the Future*, describes the future vision of S&T for the country, and identifies eight policy directions. The 2045 vision aims to create linkages between the *Going Together Hopeful Korea 2030* vision from 2006 and the *2045 Vision for Innovative, Inclusive Growth* from 2019. These documents, which have a timeframe of between 25 and 30 years into the future, aim to provide long-term guidance for S&T mid-term strategies and plans, which are established every five years, particularly the S&T Basic Plans. *Innovate Korea 2045* sets out the vision for the long-term desired orientations for Korean society and identifies the S&T challenges that must be addressed to realise the vision. It provides eight policy directions that have informed the latest S&T Basic Plan.

The presidential agenda: The presidential agenda provides broad directions for future reforms across government. The 2022 presidential programme agenda includes 120 policy tasks – nine of which are assigned to MSIT and 12 to MOTIE and cover a variety of innovation and industrial policy priorities.

The S&T Basic Plans: The five-year plans are historically the most authoritative document for STI policy in Korea, and they provide more concrete actions in terms of how the government plans to steer STI policy for relevant ministries and agencies. The Basic Plans are structured along broad strategic directions and corresponding policy areas increasingly focused on solving societal issues, in line with the overall vision to build a more inclusive STI system. In contrast with most western STI strategies that only include objectives and targets, followed by an action plan in the best case, the wide orientations of the Basic Plans are complemented with more or less concrete agendas to be implemented by different policy bodies across

the whole government structure. Therefore, the Korean Basic Plans are both a mid-term strategic framework and a mid-term action plan.

The 5th Science and Technology Basic Plan 2023-27 was released at the end of 2022, which reflects the directions from the 2045 vision and presidential programme agenda. The 2023-27 plan has three main themes, which will be discussed later in the context of Korea's mission-oriented approach:

- transforming the S&T system for qualitative growth;
- strengthening capacity-building ecosystems for open innovation; and
- resolving national challenges and preparing for the future based on S&T.

STI strategies and plans in specific sectors and areas: While the Basic Plan is meant to be the highest-level strategic document for STI across the Korean government, each ministry with significant R&D activities (i.e., MSIT, MOTIE, and sectoral ministries) also develops a mid-term action plan for MSIT to review (this process is discussed below under the policy co-ordination section). These ministries also develop annual and multi-year broad plans and targeted strategies that have historically prioritised certain industries and technologies. These are sometimes identified under the five-year plans, which adds to their legitimacy. Examples of recent subject-specific strategies include the 2050 Carbon Neutrality Strategy (see Box 3.1. for more information);¹⁴ Polar Science Future Development Strategy;¹⁵ Digital New Deal Action Plan;¹⁶ the Comprehensive Plan for Therapeutic Agriculture Research, Development and Promotion (2022-26);¹⁷ the 4th Basic Plan for Biotechnology Promotion (2023-32);¹⁸ and the 4th National Nanotechnology Roadmap (2023-32).¹⁹ These strategies outline areas for government action, and often include technological roadmaps that provide innovators with a general direction for the types of technological capabilities the government wants them to prioritise.

Many legal acts set out the roles and rules guiding these institutions at all levels. Some of the main ones include:

- The Framework Act on Science and Technology (2001), which establishes the STI governance processes, from strategic orientation to co-ordination, programming, budgeting, monitoring and evaluation;²⁰
- The Industrial Technology Innovation Promotion Act (2006), which was revised in 2020, sets all competencies and procedures to promote innovation of industrial technology and develop the necessary infrastructure;²¹
- The National R&D Innovation Act (2021), which aims to rein in and homogenise the structures and processes of the R&D management agencies and form an “autonomous and responsible” environment for supporting R&D activities;²² and
- The Act on the Performance Evaluation and Management of National Research and Development Projects (2005), which governs R&D programme and institution evaluations.²³

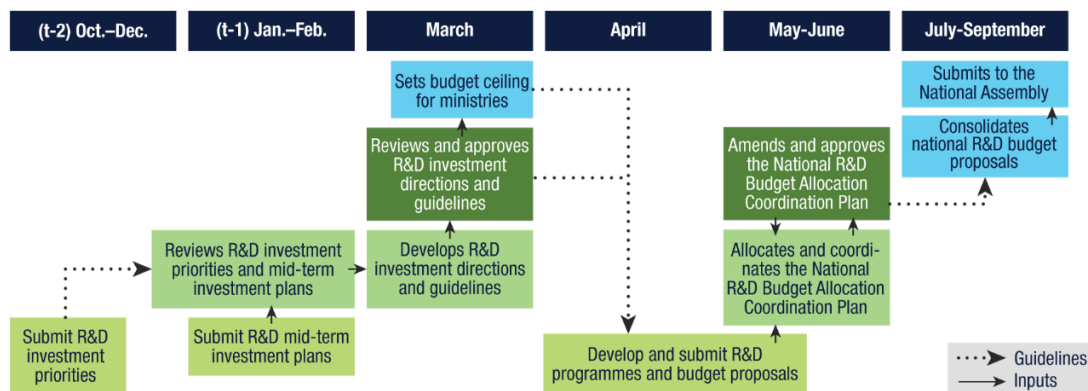
The budget process is also an important aspect of agenda setting. A comprehensive process is in place to determine the overall R&D budget and allocate it among the different ministries while adhering to the budget threshold set by the MOEF for each ministry.

As shown in Figure A.2, the annual budgeting and programming process starts in the fourth quarter of the year $t-2$ (year minus two) with ministries submitting their informal and broad priorities for the year after next (in two years). Based on these inputs, the Science, Technology and Innovation Office (hereafter, the STI Office) provides guidance on the year $t-1$ (year minus one) for the next year's programmes and ministries' budget requests (called the R&D investment directions). The R&D investment directions are also used to

inform the overall budgetary indications. After R&D budget ceilings for each ministry are set by the MOEF (usually at the end of April), based on these maximum budget allocations, all ministries develop their R&D programme proposals (until the end of May) in consultation with their own committees (for instance, the General Energy Committee that advises MOTIE) and networks of stakeholders for the next year. These proposals focus on whether to continue funding existing multi-annual programmes, and sometimes establishing new programmes.

All R&D programmes are submitted for review to the STI Office. Both the content and the budget requests of these programmes are reviewed before being approved by the Deliberative Council of the PACST. The revised proposals are transmitted in the form of the National R&D Budget Allocation and Co-ordination Plan to the MOEF for the final decision on the programmes and their respective budgets. Once aggregated, the MOEF transmits the government's annual R&D budget to the National Assembly, which approves it 30 days before the beginning of December of the year $t-1$.

Figure A.2. Overview of Korea's annual budgeting and programming process



Source: OECD (2023^[35]), *OECD Reviews of Innovation Policy: Korea 2023*, OECD Reviews of Innovation Policy, OECD Publishing, Paris, <https://doi.org/10.1787/bdcf9685-en>.

To provide more direction, the first five-year R&D Investment Strategy 2023-27 sets indicative five-year R&D budgetary and spending targets for investments in specific areas – although it does not include budget commitments, which are only determined yearly. It includes the overall budget target of 170 trillion won (~120 billion euros). The strategy is more actionable than the Basic Plan and aims to facilitate the five-year programming by the sectoral ministries and support the consistency check by the STI Office.

Strategic policy intelligence, consultations and monitoring

Strategic policy intelligence is an important aspect of Korean policymaking, and the country is broadly recognised as being at the forefront of foresight and technology forecasting. It has not only devised new methodologies and tools but also ensured that their results are used to inform strategic decisions, notably in the context of the S&T Basic Plan development process. Foresight systematically informs whole-of-government and sectoral strategies and plans – including the mission-oriented policies discussed in Chapter 4. These exercises create opportunities to debate and make better policy decisions, especially where these involve societal judgements or that have a systemic character. Foresight is particularly well connected to the process of developing the Basic Plan.

The government has established and mainstreamed formal processes to ensure the linkages between STI foresight and mid- to long-term strategy. The first major foresight exercise was run in 1994, and so far, six exercises have been undertaken. Since then, foresight has become more systematic and formally integrated into strategic planning, as a formal foresight exercise is mandatory every five years according to the Framework Act on Science and Technology. This foresight work plays an important role in informing the Basic Plan about recent changes in the domestic and international environment, devising scenarios about future technologies and deriving options for future mid-term and long-term evolution of Korean society and implications for STI priorities.

While the process changes for each five-year foresight exercise, the KISTEP is in charge of the general management of the entire process and provides support to a number of foresight committees. The 6th Foresight Exercise was implemented following a two-stage process:

1. The Future Scanning Committee and the Future Technology Committee operated a series of steps, including trend analysis using, notably, the STEEP (Social, Technological, Economic, Environmental, and Political) analysis and domestic and foreign future outlook reports. The exercise identified five megatrends, 12 trends and 62 major issues expected to considerably impact Korea's socio-economic prospects up to 2045.
2. In a second phase, the Future Technology Committee identified 241 technologies expected to emerge by 2045. The committees then conducted a two-round Delphi survey addressed to S&T experts to assess the potential impact and the corresponding societal and policy tasks to perform. Finally, based on the previous survey results, a tipping point analysis was conducted among members of the Future Technology Committee to identify 15 major future innovative technologies expected to have a considerable socio-economic impact when they are diffused in society in the future. The 6th Foresight exercise concluded that all 15 technologies are expected to reach their tipping point first in the United States.

Although the foresight exercises increasingly take into account societal aspects, the exercise remains centred on technology forecasting and culminates in a set of technological guidelines.

Stakeholder consultations are also rooted in the Korean policymaking process through several formal channels. For instance, to set the 2045 long-term vision, the STI Office conducted consultations with the line ministries and a dedicated ad hoc committee, the Future Strategy Committee 2045, which was composed of around 20 experts from industry, academia and research institutes.

Developing the S&T Basic Plan also involves several consultation stages with a broad range of communities. The process is again led by the STI Office, which begins before the presidential election with preparatory work by institutions such as the KISTEP and the Science and Technology Policy Institute (STEPI). The development of the Basic Plan also includes an assessment of the former plan and the provision of analytical support for S&T planning, notably through an analysis of changes in the overall context; a policy demand survey among STI institutions; and various forms of consultations with the main STI communities (universities and various types of research institutes, industry, and intermediary organisations, etc).

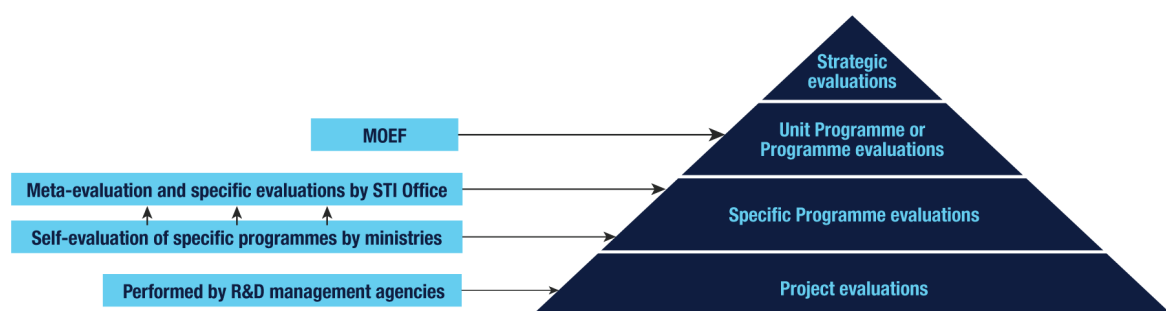
The Planning Committee of the S&T Basic Plan, a specific ad hoc committee, is formed every five years by MSIT to support the ministry in planning the process for developing the Basic Plan. This committee supervises an elaborated planning structure, while the STI Office is responsible for drafting the plan. In the case of the 5th S&T Basic Plan, a Co-ordination Committee of approximately ten members appointed by the Minister of Science and ICT develops the overall direction and collects feedback on the plan. Four

Divisional Committees (all composed of approximately 11 experts) develop accompanying agendas for, as well as implementation strategies and detailed action initiatives. The four Divisional Committees are supported by several sub-committees focusing on specific issues (e.g., digital transformation or regional decline). Once a first draft of the plan is developed, the PACST²⁴ approves the plan prepared by the MSIT and other ministries.

A system of laws, guidelines and procedures frames the **monitoring and evaluation of programmes and projects**. Ministries develop an evaluation strategy plan for each programme and monitor its implementation. Monitoring and evaluation against goals and objectives is also done regularly by the STI Office's Performance Evaluation Policy Bureau, as defined by law. It has a few key functions in this regard, including performance evaluations of national R&D programmes, performance evaluations of R&D projects, ex post meta-evaluations of national R&D programmes and special programmes, and meta-evaluation of GRIs and National Research Council for Science and Technology (NST).

In 2021, the Act on the Performance Evaluation and Management of National Research and Development Projects, originally enacted in 2005, was significantly revised. Importantly, the autonomy and responsibility of sectoral ministries for programme evaluation were strengthened, while the role of the STI Office focused on monitoring rather than evaluation. The recent reforms also intended to focus monitoring and evaluation processes above the level of standalone projects to apprehend broader strategic considerations, which has been previously identified as a limitation of the system, and will be even more important for MOIPs (i.e., exploring how multiple projects, programmes and policies benefit the sum of their parts instead of the effective implementation of stand-alone activities).

Figure A.3. Korea's different types of evaluation by programme



Source: OECD (2023^[35]), *OECD Reviews of Innovation Policy: Korea 2023*, OECD Reviews of Innovation Policy, OECD Publishing, Paris, <https://doi.org/10.1787/bdcf9685-en>.

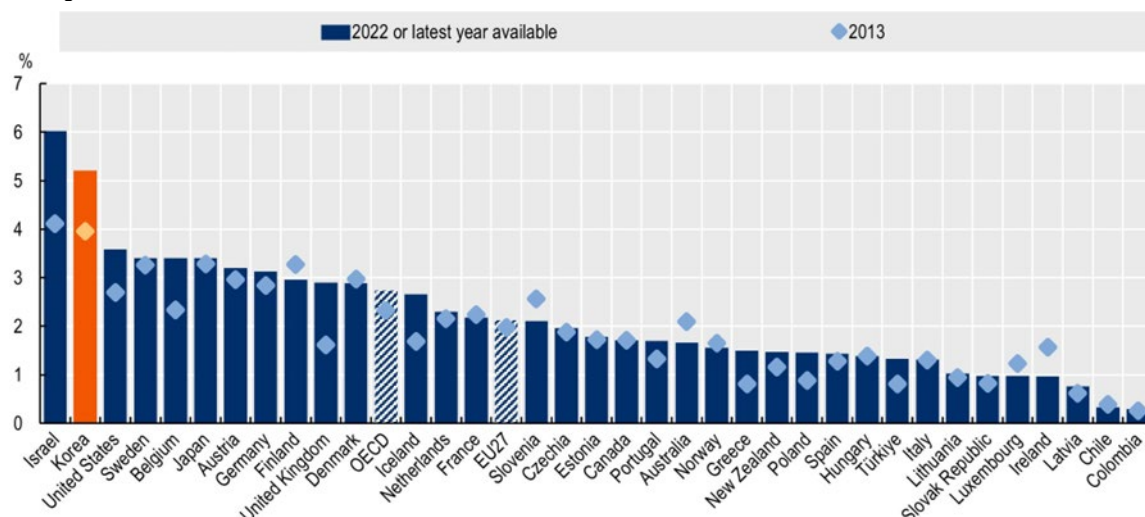
An evaluation of the outcomes and weaknesses of the latest Basic Plan is also conducted every five years by MSIT with support from relevant organisations, such as KISTEP and STEPI, as part of developing the next plan.

STI to address societal challenges

Korea has the second-highest R&D intensity among OECD countries. Its gross domestic expenditure on R&D (GERD) represented 5.21% of GDP in 2022. An exceptionally high business expenditure on R&D (BERD) accounts for the vast majority of Korea's GERD. The Korean BERD is the second highest after Israel among OECD countries (4.14% of GDP in 2022).

Figure A.4. Gross domestic expenditures on R&D (GERD), (2022 and 2013)

Percentage of GDP



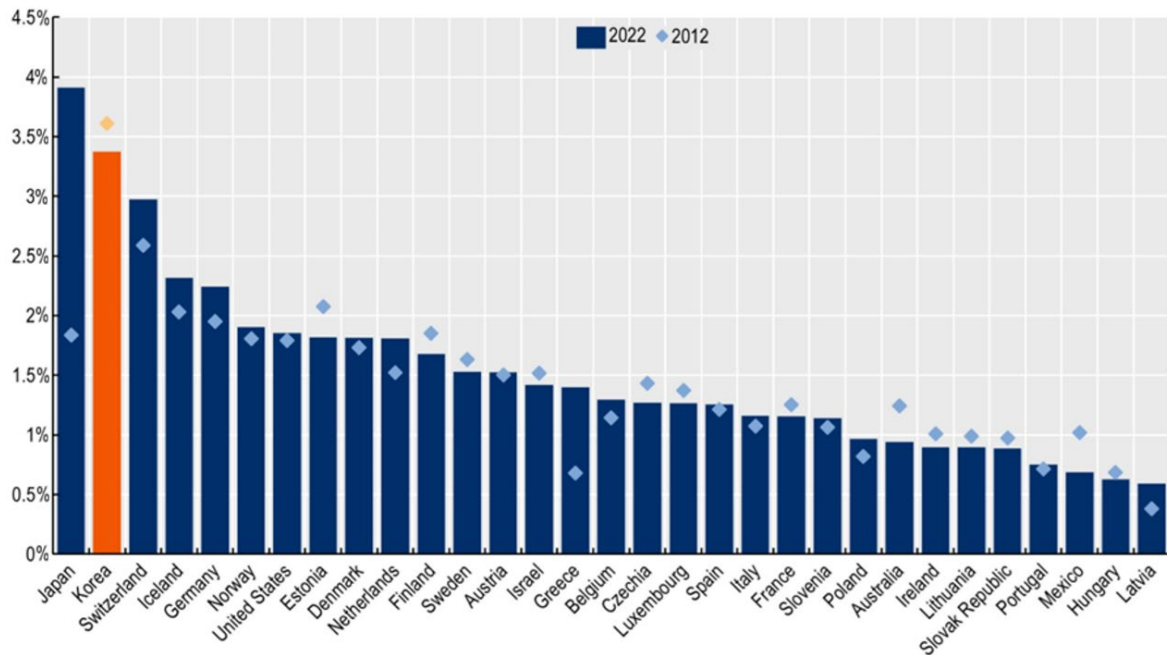
Note: Provisional values for Austria, Canada, Czechia, United Kingdom. Estimation values for Chile, Colombia Japan, European Union 27 and OECD 2022 correspond to 2021 data for Belgium, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Israel, Italy, Latvia, Lithuania, Luxembourg, Mexico, Netherlands, New Zealand, Norway, Poland, Portugal, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, United Kingdom, European Union 27 and OECD and 2020 for Chile and Colombia

Source: OECD (2024^[20]), Main Science and Technology Indicators (database), <https://www.oecd.org/en/data/datasets/main-science-and-technology-indicators.html>.

Korean government budget allocations for R&D (GBARD) are also among the highest amongst OECD countries, as Korea ranks fourth after the United States, Japan and Germany. For GBARD as a percentage of general government expenditures, Korea ranks 2nd, only behind Japan (3.37% of GBARD in 2022). Korea also ranks 1st amongst OECD member countries for GERD financed by government as a percentage of GDP (1.18% in 2022), which shows how highly the government values STI among all other areas of government interventions.

Figure A.5. Government Budget Allocations for R&D (GBARD), (2022 and 2012)

Percentage of general government expenditure

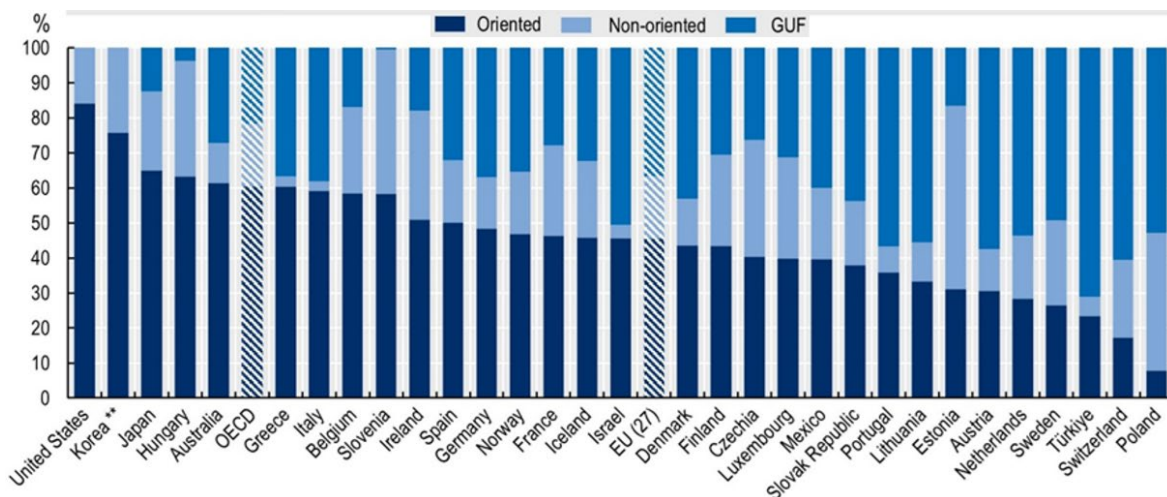


Source: Calculations based on (OECD, 2024^[36]), "Research and Development Statistics: Gross domestic expenditure on R-D by sector of performance and socio-economic objective", OECD Science, Technology and R&D Statistics (database), and OECD, Annual government non-financial accounts and key indicators (Expenditure).

A large portion of Korea's government appropriations dedicated to R&D is thematically oriented (Figure A.6). Oriented research includes research reported by national authorities as contributing to specific socio-economic objectives.

Figure A.6. Oriented and non-oriented civil GBARD, (2022)

Percentage of civil GBARD



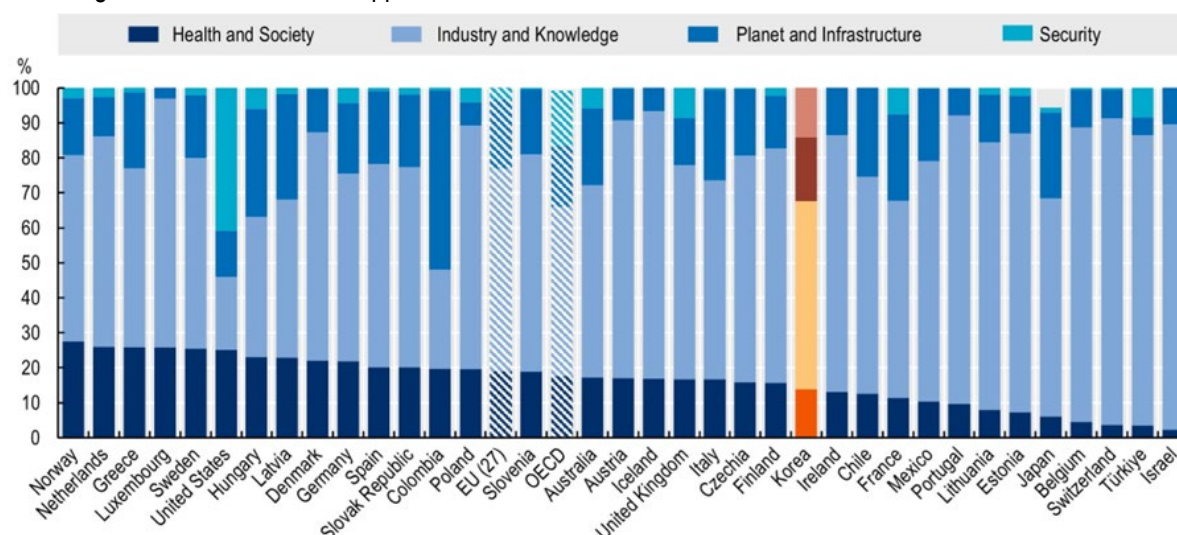
Note: Non-oriented research for Korea includes 'other' category. Definitions differ also from the Frascati manual for Australia, Japan and Mexico. Estimated value for Denmark, OECD, EU(27), Sweden.

Source: OECD (2024^[20]), Main Science and Technology Indicators (database), <https://www.oecd.org/en/data/datasets/main-science-and-technology-indicators.html>.

In Korea, public funds for oriented research accounted for 76% of the total (direct) funding for R&D in 2022, in comparison to 84% in the United States, 65% in Japan, and 48% in Germany. Non-oriented research includes both research for the general advancement of knowledge and institutional ('block') funding. This is notably related to one important feature of the Korean research system: the low level of block funding. The dominance of project-based financing makes for limited autonomy and short-term focus. Block funding for research has been introduced for some universities and government-funded research institutes, and this approach is growing. A large part of funds still depends on project-based funding, which might hinder long-term research agendas and building large research teams difficult.

Figure A.7. R&D budgets by SDG-related category, (2021 or latest available data)

Percentage of total Government support



Note: This is an experimental indicator. GBARD statistics available by Socio-Economic Objectives (SEOs) are presented in clusters that are thematically related to groups of Sustainable Development Goals (SDGs). Four indicators (of which this one) have been derived: 'Health and Society' includes 'Education', 'Culture, recreation, religion and mass media', 'Political and social systems, structures and processes', 'Health' and 'General advancement of knowledge: R&D related to Medical and health sciences' (Socio-Economic Objectives (SEOs) 09 to 11, 07, 123 and 133); 'Industry and Knowledge' includes 'Industrial production and technology', 'General advancement of knowledge: R&D related to Natural sciences, Engineering and technology, Social sciences, and Humanities and the arts' (SEO 06, 121, 122, 125, 126, 131, 132, 135, 136) as well as indirect government support through R&D tax incentives; 'Planet and Infrastructure' includes 'Exploration and exploitation of the Earth', 'Exploration and exploitation of space', 'Transport, telecommunication and other infrastructures', 'Environment', 'Energy', 'Agriculture', and 'General advancement of knowledge: R&D related to Agricultural and veterinary sciences' (SEO 01, 02, 03, 04, 05, 08, 124 and 134); and 'Security' includes Defence (SEO 14). For some countries, all 'General advancement of knowledge' sub-categories are missing. In this case, all 'General advancement of knowledge' is included in 'Industry and Knowledge' and excluded from 'Health and Society' and 'Planet and Infrastructure'. For some countries, this exception applies only for a limited period of time (e.g., where detailed historical data are unavailable). More details on data coverage can be found in the underlying R&D dataset used for constructing this indicator: see OECD, Research and Development Statistics (RDS) Database, GBARD dataset, http://stats.oecd.org/Index.aspx?DataSetCode=GBARD_NABS2007.

2021 data corresponds to 2020 for Colombia, Israel and United States

Source: OECD (2024^[20]), Main Science and Technology Indicators (database), <https://www.oecd.org/en/data/datasets/main-science-and-technology-indicators.html>.

The allocation of public funds is rooted in the historical context of a country, including the pace of development, industrial structure, and government policies. Korea dedicates a relatively small proportion

of its government R&D budget to projects and activities related to health and society issues, due to its historical need to invest more in defence areas. In 2021, 14% of R&D funds were allocated to health and society issues, compared to 28% in Norway, 25% in the United States and 22% in Germany (Figure A.7). As in most countries, it is experiencing an upward trend (it was 9% in 2012). A higher share in comparison to other countries is allocated to security-related R&D (14%, second to the United States). As in almost all countries, most of Korea's R&D resources are directed towards industry and knowledge (54%). According to research by KISTEP, the share of the Korean R&D budget dedicated to societal issues is rising (a 10% increase in 2021) but, according to analysis from KISTEP, this remains low, representing only 5.8% of the total R&D budget (KISTEP, 2021^[37]).

Policy co-ordination

Korea has a well-established three-tiered governance structure composed of dedicated institutions with clear legal mandates to perform the essential functions necessary to steer, co-ordinate and implement STI policies. As is typical of advanced economies, the Korean STI system is composed of three main levels of governance:

1. **Strategic orientation/agenda setting:** the level of the executive and legislative branches of government where the strategic framework guiding STI activities is developed.
2. **Co-ordination and programming:** the level of the individual ministries or administrations.
3. **Policy implementation:** the level of funding agencies (also known as intermediary or management agencies) which implement policy on behalf of the ministries.

The organisations that perform R&D or innovation (universities, public research institutes, companies and others, such as hospitals) form a fourth level that receives funding and conducts activities to generate impact.

Horizontal co-ordination by the President

The president is a powerful actor elected for a single and non-renewable five-year term. Members of the National Assembly serve for four years. The President appoints the Prime Minister as his/her deputy. The President appoints the other ministers, and the Prime Minister runs the Prime Minister's Office (Chancellery).

The PACST, currently chaired by the President, was created in 1991 to provide strategic, medium- to long-term advice on S&T, and co-ordinates R&D policy across the ministries (except defence, which is treated separately from the rest of the system). It focuses on national tasks that require comprehensive co-ordination across ministries and relevant stakeholders, and that require a decision from the highest level of the S&T community, namely, the President. The PACST has an advisory body, which provides high-level strategic advice to the President, and a deliberative body, which focuses on planning, budget formation and monitoring of R&D programmes. Both are chaired by the President of Korea and the PACST Vice-Chair. All ministries are expected by law to implement the PACST's decisions. At the same time, because Korea has a number of high-level councils, there is inevitably some overlap between the PACST and other councils. The PACST is supported by the PACST Support Group (about 25 staff), GRIs and other agencies that have significant analytical capabilities. This group supports both the advisory and deliberative bodies. It is comprised of public officials from relevant S&T ministries, and experts from GRIs and agencies.

In addition, the current President appointed a Senior Secretary for Science and Technology as a senior secretary to the President as part of a reorganisation of the Office of National Policy in 2024. This Senior

Secretary assists the President and co-ordinates the country's science and technology-related policies. This position had existed under previous governments, but not provided with the level of authority a senior secretary has. The Senior Secretary's office has a Secretary for R&D Policy, a Secretary for AI and Digital Transformation, and a Secretary for Bioscience & Biotechnology who help co-ordinate all science and technology policies related to their areas. In addition, the Secretary for Climate and Environment is also under the Senior Secretary. This secretary position also existed in the past but was recently moved under the STI senior secretary.

Horizontal co-ordination by the Ministry of Science and ICT

Many ministries fund R&D in Korea but the MSIT is the lead ministry in terms of funding volume and when it comes to co-ordinating R&D activities. R&D programmes related to social science and humanities are co-ordinated just by the MOEF, while the rest of the budgeting authority for R&D has been partially delegated to the MSIT to co-ordinate – representing 26 ministries and departments in other disciplines, which accounted for about 80% of the total R&D budget in 2022.

Table A.1. Internal structure and functions of the Science, Technology and Innovation Office

Bureaus	Divisions	Functions
Science and Technology Policy Bureau	Science and Technology Policy Division	• Develops S&T policy agenda • Establishes S&T Basic Plans and annual action plans and monitors their implementation • Reviews, analyses and co-ordinates mid- to long-term plans (on R&D) of sectoral ministries • Supports the operations of PACST • Establishes policies bridging R&D budgeting and evaluation • Facilitates inter-ministerial co-ordination • Establishes comprehensive plans and annual implementation plans for social problem-solving R&D • Develops and co-ordinates critical and emerging technology policy
	Science and Technology Strategy Division	
	Science and Technology Policy Co-ordination Division	
	Critical and Emerging Technology Policy Division	
R&D Investment Co-ordination Bureau	R&D Budget Co-ordination Division	• Establishes annual investment direction and guidelines • Establishes mid- to long-term investment plans • Oversees matters concerning: ○ Allocation and co-ordination of national R&D programmes and budget of GRIs ○ Establishment and co-ordination of investment strategies for national R&D programmes among R&D actors (industry, academia, GRIs) ○ Improvement of investment efficiency in national R&D programmes, such as co-ordination of overlapping programmes and structuring and restructuring by industry fields • Oversees the establishment of strategies, budget allocation and co-ordination and/or investment directions for specific disciplines/industry fields/technology areas (e.g., space, energy, and environment, ICT and convergence technologies)
	R&D Investment Planning Division	
	Public and Energy R&D Budget Co-ordination Division	
	Advanced Manufacturing and ICT R&D Budget Co-ordination Division	
	Biotechnology R&D Budget Co-ordination Division	
Performance Evaluation Policy Bureau	Performance Evaluation Policy Division	• Establishes and implements plans and actions related to: ○ Performance evaluation of national R&D programmes ○ Performance management and diffusion of outputs of national R&D programmes ○ Performance evaluation of R&D projects • Provides institutional support for the creation of knowledge ecosystems • Performs (ex post) meta-evaluation of national R&D programmes and special programmes • Performs meta-evaluation of GRIs and National Research Council for Science and Technology (NST)
	R&D Assessment and Evaluation Division	
	Research System Division	
	R&D Ethics and Researchers' Rights Protection Division	
	S&T Information and Analysis Division	
	R&D Feasibility Study Team	

		• Performs pre-feasibility tests • Oversees matters concerning: ○ Foresight exercises and strategic policy intelligence ○ Improvements in administrative processes in national R&D management (including agencies, regulations, etc.) following the adoption of the National R&D Innovation Act
--	--	--

Note: The list of functions is not exhaustive.

Source: OECD (2023), *OECD Reviews of Innovation Policy: Korea 2023*, OECD Reviews of Innovation Policy, OECD Publishing, Paris, <https://doi.org/10.1787/bdcf9685-en>.

An important actor to co-ordinate this work for MSIT is the STI Office Vice Minister that supports MSIT's co-ordination function horizontally across the entire government structure. The STI Office plays the role that is often referred to as the "STI control tower" and has a powerful mandate, defined by law, that pertains to the three levels: it leads key processes to set orientations, co-ordinate plans and budgets, and monitors and evaluates policy implementation. It is staffed with specific subject matter experts and has an extensive set of more specific responsibilities, outlined in Table A.1.

The STI Office and the PACST oversee multiple cross-government co-ordination processes in close collaboration. The STI Office often develops S&T strategies or reviews work from sectoral ministries, which is then validated or given formal approval by the PACST.

For instance, sectoral ministries submit their annual R&D action plans to the STI Office for review. The STI Office integrates these inputs into an annual Basic Plan, which it submits to PACST's Steering Committee under the Deliberative Council for review. This plan is both a monitoring (activities of the past year) and an action plan (activities planned for the next year) covering the whole government. It does not include budget figures. After identifying overlaps, the STI Office encouraged the ministries concerned to reorganise their programmes. In addition, some sectoral ministries develop annual plans and submit them to the PACST's Deliberative Council via the STI Office.

In a similar process, each relevant ministry also submits a mid-term action plan to the STI Office, which reviews them to ensure they align with the Basic Plan and do not overlap with other ministries' plans. The STI Office can interact directly with the ministries when changes are needed. At the end of the review process, MSIT delivers a list of recommendations that may require the ministries to amend their plans and take them into account in their annual action plans. Every year, the results of this review process are submitted to the PACST Deliberative Council for validation and shared with the MOEF and other ministries for information.

The STI Office annually reviews R&D-specific programmes with support from eight expert committees comprised of about 20 scientific and technological experts from the public and private sectors in specific areas (energy, ICT, and convergence technologies, among others). These experts assess the technical soundness and feasibility of the different programmes submitted by the ministries. These assessments are complemented by the STI Office's analysis of the coherence of the different programmes, both between themselves and with the priorities outlined in the Basic Plan. In addition, the STI Office can provide additional budgets to some programmes if deemed necessary, although this is relatively rare.

The STI Office also conducts 'pre-feasibility tests' – a review process that is a pre-condition for the launch of large programmes with total expenditures over 50 billion won (~34 million euros) and public support of more than 30 billion (~20 million euros). The pre-feasibility test is very stringent, as only 40% of submitted large programmes pass a pre-test, and 52% of the programmes that pass the pre-test succeed in passing a main test. The first stage pre-test is based on four criteria: 1) necessity and urgency considering recent S&T developments; 2) appropriateness to be funded by the national R&D budget; 3) novelty, yet having

foreseen linkages with the existing programme; and 4) specificity of the programme plan. If the programme successfully passes the first stage, the main test is then carried out in about seven months. The assessment of the main test uses three high-level criteria with mid-level indicators:

- **S&T feasibility:** Three mid-level indicators evaluate the adequacy of the “context of the identified problem/issue”, the “programme goal”, and the “specific tasks and implementation strategies”.
- **Policy relevance:** Two basic mid-level indicators examine the “consistency with existing policies (namely, high-level strategies)” and “risk factors (including financial and legal aspects)”. If the programme concerns wider matters outside the conventional STI policy arena, such as balanced regional development, job creation or safety assessment, it may be selectively scrutinised with one additional “special” indicator.
- **Economic feasibility:** Cost-effectiveness analysis (E/C; if quantification is not possible but at least comparative alternatives exist) or cost-benefit analysis (B/C; if the programme's effect can be quantified) is performed, and the results of the analyses are presented along with the total project cost estimate.

The government has abolished the pre-feasibility test and replaced it with a more streamlined version for large projects (MSIT and MOEF, 2024^[38]).

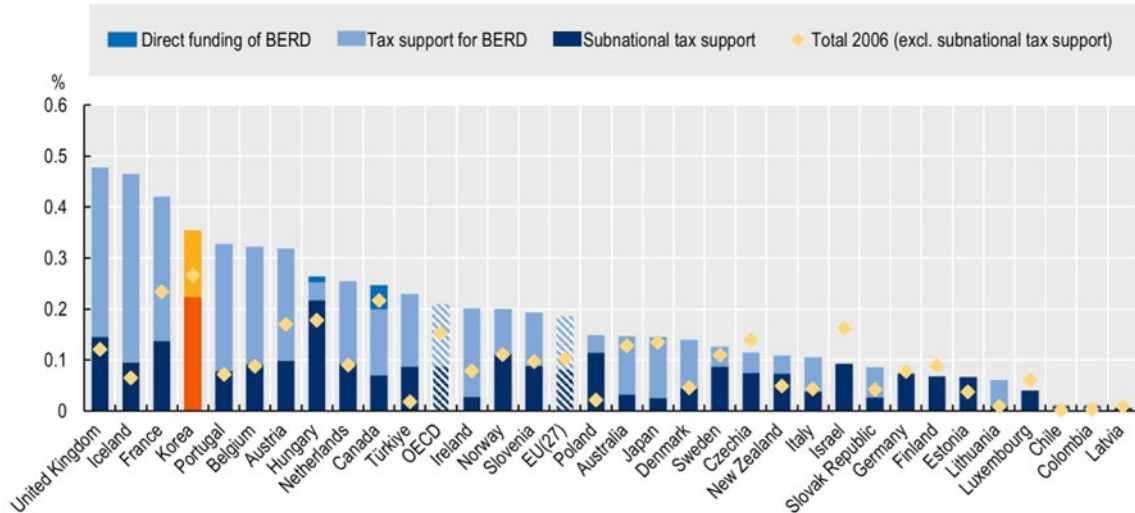
STI Policy implementation

Overview of Korean STI support

Korea not only has a very high BERD, but it is also among the OECD countries that provide the largest level of total government support to BERD as a percentage of GDP, at a rate of 0.35% of GDP in 2021 (Figure A.8). Most of this public support for business R&D is directed at SMEs. In 2022, the government financed 16.0% of all R&D conducted by SMEs. In contrast, the government financed only 2.13% of the R&D expenditures of large firms (MSIT and KISTEP, 2024^[39]). This heavy focus on SMEs in public support for business R&D applies not only to direct funding but also to tax support. Specifically, tax credits for large firms are capped at 2% of R&D spending, while there is no ceiling for the R&D tax credits for SMEs. As a result, the implied tax subsidy rate for profit-making SMEs was 26% in 2020, in contrast to 2% for large firms. Nonetheless, due to the dominant role of large firms in R&D spending in Korea, the share of the total amount of R&D tax credits given to SMEs among all R&D tax credits was only 40% in 2018 (OECD, 2023^[1]).

Figure A.8. Direct government funding and government tax support for business R&D (2021 or nearest year and 2006 or closest available data)

Percentage of GDP



Note: For general and country-specific notes on the estimates of government tax relief for R&D expenditures (GTARD), please see <http://www.oecd.org/sti/rd-tax-stats-gtard-ts-notes.pdf>. Estimation for OECD and EU27 aggregates.

Source: (OECD, 2024^[21])

Korea is one of the only countries that has directionality embedded into its tax credits for R&D. It has an enhanced tax credit rate that applies in the case of R&D aimed at ‘new growth and key enabling technologies’ (235 technologies in 12 areas) and seven national strategy technology areas (Appelt and Galindo-Rueda, 2023^[40]). For R&D investments that are aligned with these priorities, the subsidy rate can be as much as 40% for large firms and 50% for SMEs (Koo, 2023^[41]).

Institutionally, the three main actors of the R&D support policy for SMEs are the Ministry of Trade, Industry and Energy, the Ministry of SMEs and Start-ups, and the Ministry of Science and ICT. In 2019, these ministries administered 42.3%, 29.3% and 8.6% of the overall government budget for R&D support directed at SMEs, respectively (Ahn, Lee and Lee, 2021^[42]). These ministries conduct a wide range of R&D support programmes that mostly focus on providing direct R&D subsidies or loans for SMEs (KISTEP, 2019^[43]).

The government’s online portal for SME support policies lists over 400 separate programmes related to technology support (Ministry of SMEs and Start-ups, 2021^[44]). STI policies are mainly implemented in the form of R&D programmes by around 15-17 dedicated R&D agencies. These are normally funding agencies, but often, due to their proximity to public and private organisations that perform R&I activities and their monitoring and evaluation role, they also provide strategic intelligence to their principals. Agencies can be mandated to allocate funding and manage projects (R&D management agencies), while others perform a range of other actions to support R&D and innovation (R&D promotion agencies). In some cases, the difference between the agency roles is unclear, with some agencies playing both. Furthermore, some agencies have the title of institutes, such as the Institute of Information and Communications Technology Planning and Evaluation (IITP), which promotes technology transfer in the IT sector and manages national R&D projects/programmes funded by MSIT.

In terms of sector performance, public research is concentrated in GRIs more than in universities, with the highest level of government expenditure on R&D (GOVERD) among OECD countries (0.49% of GDP). Higher education expenditure on R&D (HERD) is also above the OECD average (0.48% vs 0.43%) (OECD, 2024^[20]).

Throughout the years, management agencies have established a set of elaborate programme management procedures throughout the whole project cycle. As in other advanced countries, these agencies are tasked with defining the precise programmes under the purview of their line ministries. Once approved and funded through the annual process described above, they develop roadmaps and calls for proposals, involving various experts and stakeholders (such as using surveys and expert committees) to organise the project selection process. The projects are then funded, managed, monitored and evaluated. They have well-established processes for performing these tasks. Within agencies, the programme directors are important actors, responsible for consultation with stakeholders and managing the portfolio of projects in their respective areas.

The programmes and the projects within are multi-annual. At the Ministry of Land, Infrastructure and Transport (MOLIT), for instance, projects are funded for three to seven years, depending on their objectives, with a majority being funded for five years. Moreover, as previously mentioned, respective R&D agencies are subject to different mandates that are formalised in a number of laws and decrees accumulated over time.

Another aspect of Korean innovation policy is the conditionality it places on its funding. Technologies that are commercialised from government-funded R&D projects are charged technology fees. This helps “maintain a virtuous cycle of technology development” by reinvesting the fees in further research and development through a quasi-dividend arrangement. The amount charged ranges for company size and type (profit or non-profit), but in general it is calculated using the following formula:

- Technology fees paid to the government = Profit from research and development results × Technology contribution (ratio determined by the head of the central administrative agency and the research and development agency through an agreement) × rate (2.5% for SMEs, 5% for medium, 10% for large) (Related Joint Ministers, 2024^[19]).

The formula reduced the rates by 50% (the above rates reflect this update) in June 2024 to ease the burden on businesses.

Inter-ministerial programmes

A number of broad and complex (“wicked”) issues require resources and capabilities that go beyond the portfolio of any single ministry or agency. In this case, co-operation between different ministries and agencies pertaining to different policy fields can prove beneficial either to cover different stages of the innovation cycle (from research to deployment), different areas of competencies (for instance, support to technological development and support to skills development) and/or different thematic areas (for instance, energy and transport).

Large programmes are one of the main mechanisms for collaboration among ministries and agencies across policy silos. The ministry-led inter-ministerial programmes tend to be longer-term and have a larger scale and scope, and therefore subject to pre-feasibility tests (KISTEP, 2019^[45]). The government established the Multi-ministerial R&D Programme Promotion Plan in 2013 to overcome the issues affecting

these types of programmes, and established the Multi-ministerial Programme Promotion Committee. This committee is chaired by the Deputy Head of the STI Office and is comprised of 11 high-level government officials and 9 private experts. It reviews and selects the R&D co-operative programmes, and deliberates on all issues regarding inter-ministerial co-operation.

The planning of multi-ministerial R&D programmes follows four steps. First, MSIT conducts two surveys to identify the needs for co-operation: the first is a bottom-up process in the form of calls for expression of interest to industry, academia and GRIs; and the second is a top-down process based on the demands from sectoral ministries. Second, MSIT supports the pre-selection of the candidate programmes, later to be approved by the Multi-ministerial Programme Committee. Third, once selected, the lead ministry assigned to each co-operative programme, in consultation with other ministries, develops a detailed programme plan that includes the division of labour among ministries. Finally, the developed plans are again submitted to the Multi-ministerial Programme Committee, which gives final approval (KISTEP, 2021^[46]).

In addition, different schemes exist to promote and enable this type of co-operative endeavour at the ministry or agency level in practice. In large cross-ministerial programmes, different ministries launch a common umbrella programme and delegate some components to their respective agencies. Other co-operation is implemented directly between agencies in joint programmes. These can be limited to ad hoc joint calls for proposals.

Annex B. Overview of the process, governance and main MOIP features of the three cases

Figure B.1. The 3rd comprehensive plan to solve social problems based on science and technology (2023-27)

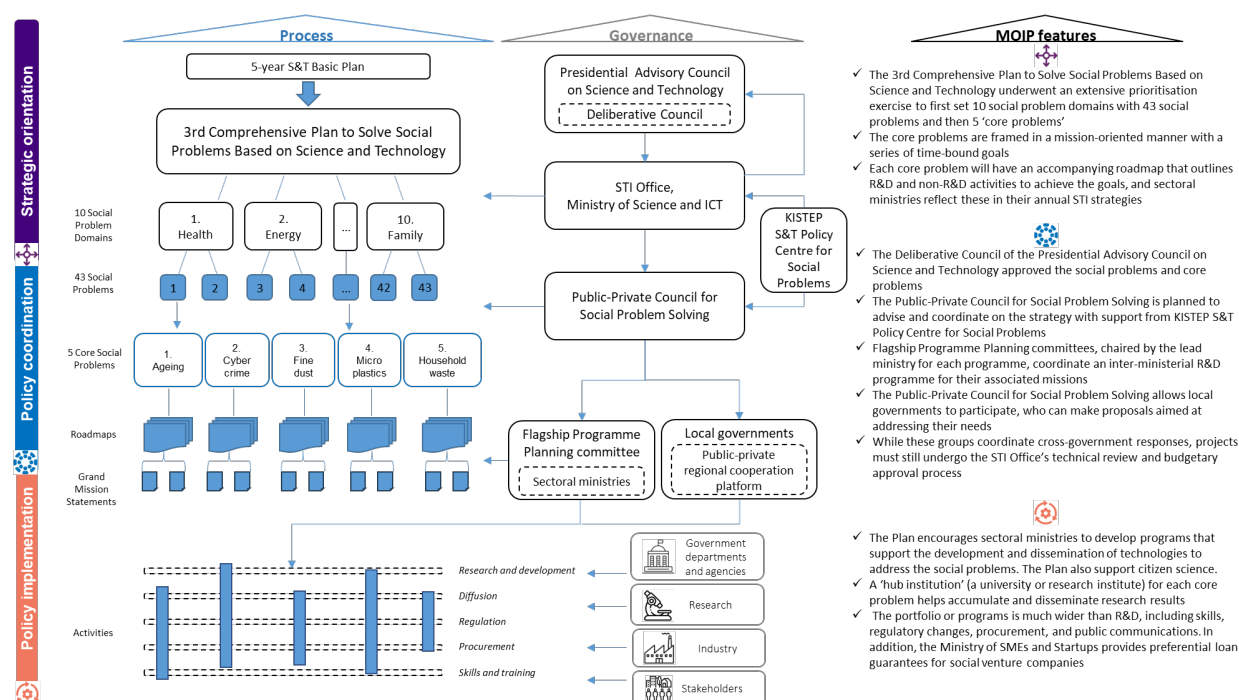


Figure B.2. The National Strategic Technology Policy

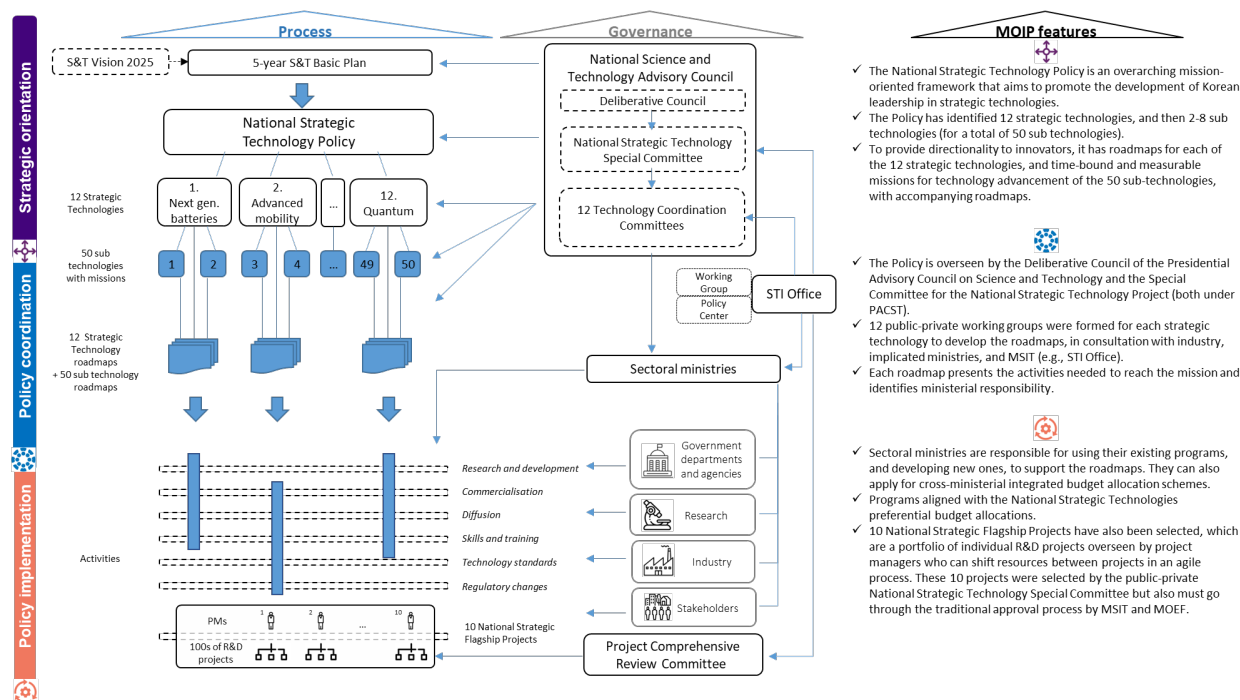
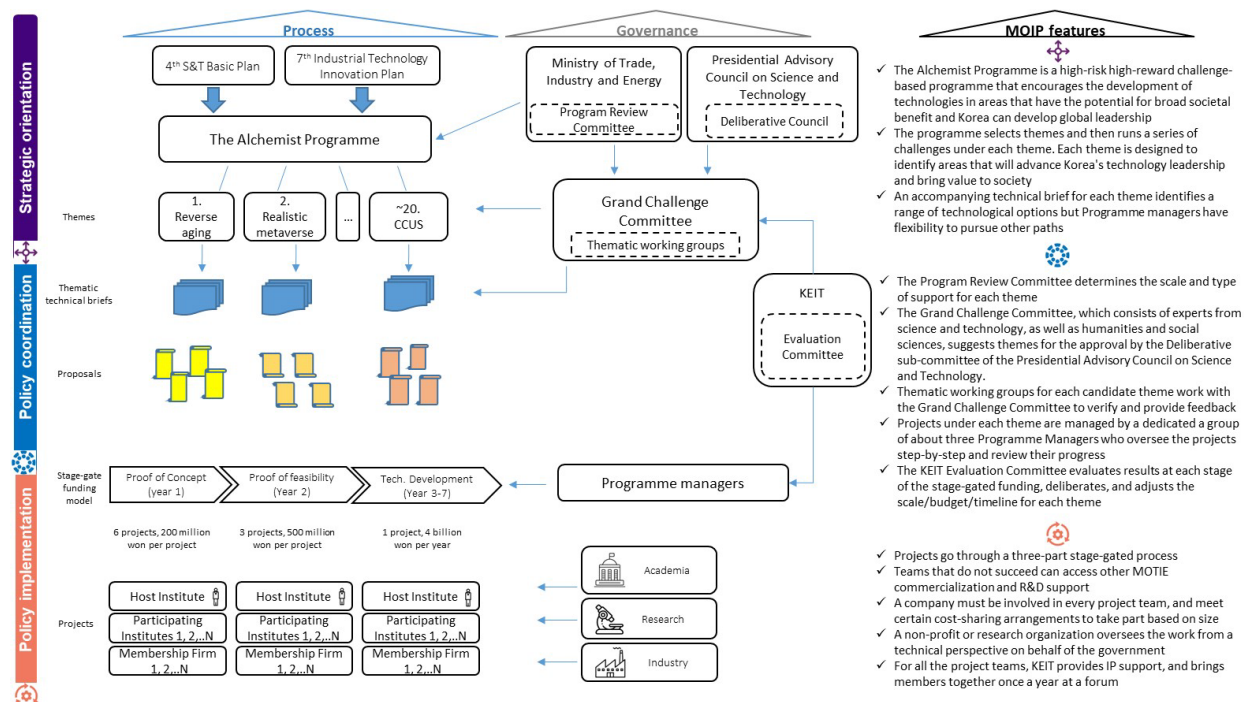


Figure B.3. The Alchemist



Annex C. 3rd Comprehensive Plan to Solve Social Problems Based on Science and Technology – List of social problems

Table C.1. 3rd Comprehensive Plan to Solve Social Problems Based on Science and Technology – List of social problems

Social problem domains	Social problems*
Health	• Chronic disease • Rare incurable disease • Addiction • Degenerative neurological disease • Mental illness and intellectual disability
Environment	• Household waste • Indoor air pollution • Water pollution • Environmental hormones • Industrial waste • Fine particle matter • Microplastic • Residential noise (NEW)
Culture and leisure	• Cultural alienation • Insufficient culture and leisure space
Life safety	• Sex crime • Food safety • Home safety • Invasion of privacy • Cybercrime (ADJUSTED) • Voice phishing (NEW)
Disaster	• Weather disaster • Chemical accident • Infectious disease • Radioactive contamination • Earthquake • Fire safety

Energy	• Power supply • Energy poverty
Housing and transportation	• Poor/old housing • Traffic congestion • Traffic safety
Family	• Domestic violence • Aging (ADJUSTED) • Low birth rate (ADJUSTED) • Single-person household alienation (NEW)
Education	• Education gap • School violence
Social integration	• Medical gap • Housing inconveniences for vulnerable people • Digital divide (ADJUSTED) • Lack of jobs (ADJUSTED) • Social polarisation and conflict (NEW)

Note: * New and adjusted social problems are in relation to the 2nd Comprehensive Plan to Solve Social Problems Based on Science and Technology

Annex D. 3rd Comprehensive Plan to Solve Social Problems Based on Science and Technology – core problems

Table D.1. 3rd Comprehensive Plan to Solve Social Problems Based on Science and Technology – core problems

Core problem	Mission areas
Ageing - Pursuing safety in daily life, including walking, and narrowing the digital gap for the elderly so that they can lead safe and active lives	Ensuring elderly life safety - Prevent safety accidents and improve convenience of living when elderly people walk and move Bridging the digital divide - To improve accessibility to digital information and digital literacy among older adults
Cybercrime - In order to relieve public anxiety caused by cybercrime, safe protection of personal data and minimization of economic and psychological damage to individuals are pursued	Preventing personal data leakage - To encrypt and share personal data when creating, collecting, and storing it Reducing personal cybercrime damage - Minimize personal damage from loss of personal financial data due to crimes such as cyber fraud, cyber financial crime, and digital sexual violence
Reducing fine particle matter	TBD
Reducing microplastics	TBD
Reducing household waste	TBD

Annex E. Reducing fine particle matter

Fine particle air pollution is a long-standing issue in Korea. It had improved from 1995 until 2002 following the introduction of environmental standards, before deteriorating again and reaching a low in 2012. The government tried to address this through the 10-year Comprehensive Plan for Air Quality Improvement (2006-2015), but a 2016 public opinion poll by the Korean Medical Association indicated that fine particle matter was still the “most feared public health risk” (Related Joint Ministers, 2016^[47]).

In response to this concern, in June 2016, the Korean Prime Minister and the Office for Government Policy Co-ordination announced the Special Countermeasures for Fine Particle Management, with the mission to improve fine particle matter (PM10) concentration to the current level in major European cities within the next 10 years (from 23 $\mu\text{g}/\text{m}^3$ in 2015 to 18 $\mu\text{g}/\text{m}^3$ in 2026). The Ministry of the Environment has led on the policies to address fine particle matter. The Strategic Council on Science and Technology included the Countermeasures for Fine Particle Management as one of the nine national strategic projects in August 2016 (Related Joint Ministers, 2016^[47]). This is an example of an inter-ministerial R&D programme, where national strategic projects are jointly delivered by actors across the innovation system, including companies, universities, research organisations, and multiple government ministries (see section 3.3 for more information on these types of programmes). Unfortunately, it did not proceed because it failed the pre-feasibility test. However, ten ministries have supported various fine particle matter projects since 2019.

The first stage of the fine particle matter project was carried out based on the Science and Technology-based Fine Dust Response Strategy from 2017 to 2020. It conducted considerable work to educate citizens on their role in addressing fine particle matter.

A new phase was announced in 2019: the Fine Dust Management Comprehensive Plan (2020-2024). It focused on a new mission to reduce the annual average concentration of ultrafine particle matter (PM2.5) by more than 35% (from a national average annual concentration of 26 $\mu\text{g}/\text{m}^3$ in 2016 to 16 $\mu\text{g}/\text{m}^3$ in 2024) (Fine Dust Special Countermeasure Committee, 2019^[48]). This was rooted in the Special Act on the Reduction and Management of Fine Dust (2019), which requires a new plan be announced every five years and was mentioned in the 4th S&T Basic Plan – which demonstrates the important emphasis the government put on it.

The Fine Dust Special Countermeasure Committee oversees the Comprehensive Plan, which consists of 36 people, half of whom are government officials and half are from the private sector or academia. The National Council on Climate and Air Quality under the President's Office functions as an advisory body, involving local governments, academia, industry, civil society, and political parties. The Office for Clean Air supports the Committee and monitors STI-related projects.

The National Air Emission Inventory and Research Centre provides policy support and surveys fine particle matter air emission. Because fine particle matter does not have borders, the Fine Dust Management Comprehensive Plan emphasises working with China and other East Asian countries to reduce their emissions. In addition, local strategies aligned with the national-level Plan provide direction at the city and regional level and help reduce fine particle matter from their local emission sources.

Fine particle matter is still a priority for Korea, as a 2nd Fine Dust Management Comprehensive Plan will be released in 2024 to cover 2025-29, and it is one of the five core problems under the CP3. Because the Fine Dust Management Comprehensive Plan includes “R&D for emissions reduction and management of fine particle pollutants and other air pollutants” as one of its main pillars, it will reflect the R&D directions

related to fine particle matter management provided in the CP3. The CP3 consulted the Ministry of the Environment for the development of the fine dust core issue roadmap.

Annex F. National Strategic Technology Policy – strategic technologies

Table F.1. National Strategic Technology Policy – strategic technologies

Strategic technologies	Sub technologies
Semiconductors and displays: Developing Low Power/High Efficiency AI Semiconductors (Power efficiency >10 TFLOPS/W)	• Highly integrated/resistance-based memory • High-performance/low-power artificial intelligence semiconductors • Power semiconductor • Semiconductor advanced packaging • Next-generation high-performance sensors • Freeform display • Inorganic light-emitting displays • Display materials/parts/equipment
Batteries: Highest performance Li-ion (energy density 350 Wh/kg), next-gen battery materials (all-solid-state, lithium metal, sodium ion, etc.)	• Lithium-ion batteries and core materials • Next-generation battery materials/cells • Next-generation battery module/system • Reuse/recycle of next-generation batteries
Advanced mobility: Lv.4 Autonomous Driving, UAM Safety Management System	• Autonomous driving system • Hydrogen/electric vehicles • Urban air transportation
Next-generation nuclear energy: World's Highest SMR Safety - Manufacturing Technology (accredited in '28)	• Small Modular Reactor • Advanced nuclear power system and waste management
Advanced biotechnology: Synthetic Biology, Digital-Bio Convergence - Advancing Biomanufacturing Efficiency	• Synthetic biology • Infectious disease vaccines/treatment • Gene/cell therapy • Digital health data analysis/utilization
Aerospace and marine: Develop 100-ton liquid engine, demonstrate lunar lander (~2032)	• Large multi-stage combustion cycle engine • Space observation/sensing • Moon landing/surface exploration • Advanced aviation gas turbine engine/parts • Ocean resource exploration
Hydrogen: Green Hydrogen Mass Production/Storage/Transportation	• Water electrolysis hydrogen production • Hydrogen storage and transportation • Hydrogen fuel cells and power generation
Cybersecurity: Increase proactive capabilities, including Zero-Trust, AI security, and more	• Data/AI Security • Digital Vulnerability Analysis/Response • Network/Cloud Security • Industrial/Virtual Convergence Security

Artificial intelligence: Streamline generative AI (50% less training data/power consumption than before)	• Efficient learning and advancement of AI infrastructure • Advanced AI modeling/decision making (cognition/judgment/inference) • Safe/trustworthy AI • Industrial use/innovation AI
Next-generation communications: Lead the 6G global market by securing standard patents and leadership in core technologies	• 5G advancement (5G-Adv) • 6G • Open-RAN • 5G-6G high-efficiency communication components • 5G-6G satellite communication
Advanced robots and manufacturing: Implement generative AI-driven movement, manipulation, and interaction (e.g., manipulating unstructured objects, overcoming adverse weather conditions)	• Robot precision control/driving parts/SW • Robot autonomous movement • High-difficulty autonomous operation • Human-robot interaction • Virtual manufacturing
Quantum Technology: Development of 9 core technologies related to quantum computers/communication/sensing (quantum processors (QPUs), quantum device connectivity, etc.)	• Quantum computing • Quantum communication • Quantum sensing

Annex G. The Alchemist – programme themes

Table G.1. The Alchemist – programme themes

Year	Themes
2024	• Ultimate Semiconductor - Semiconductor devices that operate reliably in environments such as high temperature/extreme temperature, ultrahigh pressure, high magnetic field, and radiation exposure, such as space, other planets, and nuclear accidents • Net generation humanoid: Humanoid that think, feel and communicate like a human - Humanoid capable of human-like movements, experiential accumulation, cognition/judgment, and communication, and measuring more than human senses (6th senses such as five senses + magnetic field) • Idea to Product: intelligent manufacturing platform with ultra-connectivity - A platform where AI can automatically design, derive optimal processes, explore and connect factories, and create products and parts without having a factory with an idea
2023	• Sustainable Beyond Plastics - Bioplastic biodegradation point control function, organic and inorganic hybrid material plastics and new material design, and material recirculation technology considering small molecule recovery • A multiverse avatar platform that shares sensations and experiences - Hyper-connected platform implementation technology that provides various social activities and experiences through virtual world realization synchronized with real space and avatars that share senses with humans • Urban Carbon Dioxide Capture and Utilization Technology - Carbon capture, storage, and conversion technologies using urban infrastructure such as buildings, roads, and energy infrastructure
2022	• Aging reversal - Identifying aging genes before cell aging / Detect/diagnose and control its expression • Aging reversal - Reversing the aging process • Ultra-realistic metaverse visualization - Implementation of ultra-realistic three-dimensional images that overcome the limits of human visual perception • Biomimetic carbon resource recycling - Carbon capture/storage/conversion that mimics life (natural) phenomena and the principles of biological structures (biomimetics) and carbon resource recovery
2020	• Brain to X - A two-way neural interface that allows you to control external devices or communicate with others using just your thoughts. • Body protective smart suit - A lightweight soft wearable suit that automatically detects harmful substances (viruses, toxic gases, fine dust, etc.) and protects the human body at all times. • Gene self-correction and healing technology - Technology to prevent or treat disease onset by monitoring and self-correcting disease genes within cells • Soft implant without immune rejection - Modular organs composed of multiple cells and replaces soft organ functions (excluding hard types (bones, teeth, joints)) without immune rejection within the body • Off-the-ground mobility - A new concept of ultra-short-distance transportation that can be moved by a person while floating on the surface. • In-situ virus detection and analysis system - Rapid detection, analysis, and innovative diagnostic system for highly transmissible and high-risk viruses such as new infectious diseases • AI based supercritical materials - Develop materials with more than the critical value (70%) of theoretical properties by integrating design of material processes and characteristics based on AI • Molecular level printer - Molecular level printing technology and products that can form two-dimensional (large-area) or three-dimensional structures at the atomic or molecular scale • CO₂ free low-cost hydrogen production - Base technology and process for mass production of CO₂-free low-cost green hydrogen with excellent economic efficiency

-
- | | |
|--|--|
| | <ul style="list-style-type: none">• Artificial eco food - Food security and environmental pollution response possible, processed food containing animal nutrients based on livestock cells |
|--|--|
-

Annex H. The Innovation Challenge Programme

The Innovation Challenge Programme was a challenge-based programme designed by the STI Office and Delivered by an internal organisation in KISTEP called the Korea Advanced Research Programme Accelerator (KARPA). The programme worked by setting themes, and then a series of research projects under each one. Project managers were in place to manage the projects in a more agile approach, winding down projects that were unsuccessful, and scaling-up those with more promise.

Each year, new themes were selected through a process that involved an extensive consultation: a ministry demand survey was conducted to identify research themes from within government; the review committee could put forward their own research themes; and the Crazy Idea Accelerator was created for researchers across the country to present their ideas. An advisory council of industry, academics and research organisations narrowed down these proposed themes, then the implicated ministries reviewed the themes and selected the first batch, and then the 'promotion team leader and division heads' decided on a second shorter list. There was then a negotiation with the selected candidate teams on the proposed research agenda. The research themes were then externally reviewed by the Innovation Challenge Project Promotion Committee (consisting of academic and industry experts) and a final research agenda was established (KISTEP, 2021^[49]).

The promotion teams ensured a diverse set of people reviewed projects to determine which ones warranted more funding. The programme sought to assign individuals to head this team who had not worked as government officials in the past and focused on individuals who had private sector technical expertise, participated in large-scale R&D projects, and had large networks domestically and internationally. The head of a promotion team was the general project manager (PM), who led the R&D project planning and managed an overall innovation challenge project (Lee et al., 2023^[50]). The Innovation Challenge Programme established three research areas to generate innovative outcomes that can bring about socio-economic transformations.

- Happy and healthy ageing
- Safe and pleasant society
- Sustainable growth of Korea's economy

Within these three areas, each year several challenges were selected based on five criteria: clarity, challenge-orientation, innovativeness, originality, and impact. The challenges were technical in nature, and their timeline ranged from 4-7 years. It launched 19 challenges from 2020-2023. The research themes fell into two types. Firstly, “exploratory” projects were funded for about four years from ideation until the proof-of-concept phase (short term). Secondly, the “package” projects funded the whole cycle from R&D to demonstration phases for a maximum of nine years (longer term) (OECD, 2023^[1]). A mid-term evaluation was conducted after the initial two years to decide whether to continue funding based on the pilot projects (Lee et al., 2023^[50]). The programme was able to select multiple research institutions to achieve the same or similar research goals. The projects funded universities, research institutes, and companies. Unlike traditional programmes, it could also fund overseas research teams (KISTEP, 2023^[51]).

The Innovation Challenge Project Promotion Team and the supervising ministry reviewed projects to determine which ones warranted more funding. Where annual evaluations are conducted for traditional programmes, the project managers would conduct regular progress inspections, in addition to stage evaluations and final evaluations at the end of a project. Evaluation also centred on looking at what was

necessary to achieve the goals put forth, rather than a traditional score/grading system. This also meant that the project managers could alter the targets, where appropriate (KISTEP, 2021^[22]).

Annex I. Main features of selected MOIP initiatives in Korea

Table I.1. Annex I. Main features of selected MOIP initiatives in Korea

	 Strategic orientation	 Policy coordination	 Policy implementation
3 rd Comprehensive Plan to Solve Social Problems Based on Science and Technology	- Concentrates government STI investment in areas that matter most towards the Korean people. - The Comprehensive Plan has identified 10 social problem domains, and 43 social problems under them, for which to focus its R&D investments on. To address these 43 social problems, Korea has identified 10 broad tasks. - Of these 43 social problems, it has selected five 'core problems' that are particularly important to Koreans. To date, for two of these core problems, it has established mission statements, another set of more specific sub priority areas (about two per problem), and specific targets for each of those sub priorities. The remaining three problems are under development.	- The Deliberative Council of the PACST approved the broad social problem domains and core problems. - The Public-Private Council for Social Problem Solving is planned to advise and coordinate on the strategy with help from MSIT (the STI Office) and KISTEP (the S&T Policy Centre for Social Problems). - Flagship Programme Planning Committees chaired by a champion ministry for each mission co-ordinate an inter-ministerial R&D programme for the mission. - Local governments can also bring their issues to the Public-Private Council for Social Problem Solving.	- The core problems have roadmaps that identify R&D and non-R&D tasks to address them. Four have been developed for two core problems and more are in development for the others. - A hub institute (a university or a government-funded research institute) for each of the core problems will identify and disseminate multi-disciplinary research. - Successful projects will receive follow on funding and support with procurement and regulations. - Citizens are encouraged to conduct 'citizen science' and learn about the solutions that get developed through a dedicated platform and targeted outreach.
National Strategic Technology Policy	- The National Strategic Technology Policy provides direction to innovators and government ministries by identifying 12 strategic technologies, developing a roadmap for each one, and prioritising funding for programmes and projects that advance the roadmaps. - For each of these 12 strategic technologies, the National Strategic Technology Policy identified 2-8 sub technologies (for a total of 50), and its strategic roadmaps present time-bound and targeted missions for the technological capabilities to achieve each one.	- The Policy is overseen by the Deliberative Council, which approved the National Strategic Technology Policy, and the individual strategic technologies. - The National Strategic Technology Policy Special Committee (chaired by vice minister in the STI Office) oversees the implementation of the Policy and promotes co-operation between relevant ministries. - Technology Co-ordination Committees have been created for each of the 12 strategic technologies to co-ordinate their work.	- The Strategic Roadmaps for each of the 12 technologies will provide the missions and enabling technologies with investment directions. - Each roadmap presents a range of activities needed to reach the mission, including technology development, commercialisation, diffusion, skills and training, regulatory changes, and technology standards. - The STI Office selected 10 flagship R&D Projects through a competitive process, which are a portfolio of individual R&D projects that need urgent investment with tangible impact in the short-term. These

			projects are overseen by project managers.
Alchemist	• The Alchemist Programme is a high-risk high-reward challenge-based programme that encourages the development of technologies in areas that have the potential for broad societal benefit. • The programme selects themes and then runs a series of challenges under each theme. Each theme is designed to identify areas that will advance Korea's technological and industrial leadership and bring value to society.	• The KEIT, one of the R&D management agencies under the MOTIE, oversees the programme and establishes a draft programme implementation plan. • The Grand Challenge Committee – which includes about 60 experts in the humanities, social and economic fields such as technology, future studies, economics, and science fiction from academia, industry and research – decides on new themes and consults with stakeholders. • A research institute is assigned to each theme for overseeing the work of each theme from a technical perspective. • Projects under each theme are managed by a dedicated group of three Project Managers (one from KEIT and two from outside) who oversee the projects step-by-step and review their progress.	• Each research theme goes through a three-stage competitive process. • Individual organisations and consortia can apply and must provide matched funding depending on the company size.

Annex J. List of interviewees

Table J.1. List of interviewees

Name	Related MOIP initiative	Affiliation
Yongseong Kang	The Alchemist	Senior Project Researcher, Challenge & Innovation Team, Korea Evaluation Institute of Industrial Technology
Jinhee Park	3 rd Comprehensive Plan to Solve Social Problems Based on Science and Technology	Director, Science and Technology Coordination Division, STI Office
Dr Jeehoon KI	3 rd Comprehensive Plan to Solve Social Problems Based on Science and Technology	Associate Research Fellow, Korea Institute of S&T Evaluation and Planning
Dr Gwi-Nam Bae	Fine particle management	Head / Principal Research Scientist, Center for FRIEND Project, Korea Institute of Science and Technology
Sojung You	Fine particle management	International Cooperation Specialist, Climate and Air Quality Policy Office, Ministry of Environment
Dr Sang-Min Park	National Strategic Technology Program	Director, R&D Investment Planning Division, STI Office
Dr Hyeok Lee		Associate Research Fellow, Office of R&D Innovation Research, Science and Technology Policy Institute
Soohong Eum		PhD candidate, University of Leiden

Note: The affiliations and titles are those valid at the time of the interview (December 2023 to June 2024).

Notes

¹ Results of this work have notably been published in (Larrue, 2021^[5]) and (OECD, 2023^[52])

² In Korean: 제 3 차 과학기술 기반 사회문제해결 종합계획 (2023-2027)

³ In Korean: 국가전략기술 프로젝트

⁴ In Korean: 알키미스트 프로젝트

⁵ In Korean: 녹색성장국가전략

⁶ In Korean: 지속 가능한 녹색사회 실현을 위한 대한민국 2050 탄소중립 전략

⁷ In Korean: 저탄소 녹색성장 기본법

⁸ In Korean: 대한민국 과학기술 미래전략 2045: 미래를 향한 도전과제와 과학기술정책의 전환

⁹ In Korean: 과학기술정보통신부

¹⁰ In Korean: 제 5 차 과학기술기본계획 2023-27

¹¹ In Korean: 산업통상자원부

¹² In Korean: 기획재정부

¹³ In Korean: 국가과학기술자문회의 기관소개

¹⁴ In Korean: 대한민국 2050 탄소중립 전략

¹⁵ In Korean: 극지과학 미래발전전략

¹⁶ In Korean: 2022 년 디지털뉴딜 실행계획

¹⁷ In Korean: 제 1 차 치유농업 연구개발 및 육성 종합계획(2022-2026)

¹⁸ In Korean: 제 4 차 생명공학육성 기본계획(2023-32)

¹⁹ In Korean: 제 4 기 국가나노기술지도(2023~32),

²⁰ In Korean: 과학기술기본법

²¹ In Korean: 산업기술혁신 촉진법

²² In Korean: 국가연구개발혁신법

²³ In Korean: 국가연구개발사업 등의 성과평가 및 성과관리에 관한 법률

²⁴ In Korean: 국가과학기술자문회의